Chapter 5

Campbell Biology, AP Edition, 12e

Model of protein shows a large and complex three-dimensional structure with variation in different regions of the molecule.

Chapter 5

The Structure and Function of Large Biological Molecules

Figure 5.1

Alcohol dehydrogenase, a protein that breaks down alcohol in the body, is shown here as a molecular model. The form of this protein that an individual possesses affects how well that person tolerates drinking alcohol. Proteins are one class of large molecules, or macromolecules.

What are the structures and functions of the four important classes of biological molecules?

Three classes are macromolecules that are polymers (long chains of monomer subunits). This is denoted by circles linked with lines to form a long chain, with one circle labeled monomer.

Carbohydrates are a source of energy and provide structural support. This is represented by a spiral made of hexagonal subunits, one of which is labeled glucose.

Proteins have a wide range of functions, such as catalyzing reactions and transporting substrates into and out of cells. A protein (alcohol dehydrogenase) is shown as a three-dimensional, coiled, ribbon-like structure where a small portion is expanded and labeled an amino acid.

Nucleic acids store genetic information and function in gene expression. An illustration of the DNA helix is shown, one portion of which is expanded and labeled a nucleotide.

The fourth class, lipids, are not polymers or macromolecules. Lipids are a group of diverse molecules that do not mix well with water. Key functions include providing energy, making up cell membranes, and acting as hormones. A phospholipid molecule is illustrated as a unit that has a head and two tails and made of globular subunits.

Key Concepts

- **5.1Macromolecules are polymers, built from monomers****
- **5.2Carbohydrates serve as fuel and building material****
- **5.3Lipids are a diverse group of hydrophobic molecules****
- **5.4Proteins include a diversity of structures, resulting in a wide range of functions****
- **5.5Nucleic acids store, transmit, and help express hereditary information****
- **5.6Genomics and proteomics have transformed biological inquiry and applications****

Advanced placement edition

All life-forms on Earth share biochemical similarities **(BIG IDEA 1)** that include large, complex molecules synthesized from common building blocks **(BIG IDEA 4).**

Study Tip

Make a visual study guide: For each class of biological molecules, draw two examples and list their structural similarities and their functions.

Important biological molecules include carbohydrates, proteins, nucleic acids, and lipids.

Get Ready for This Chapter

In this chapter, you will learn about the structures and functions of the four classes of large biological molecules. Before beginning this chapter, you should be able to do the following:

- Explain the bonding behavior of a carbon atom (see Concept 4.2, section entitled “The Formation of Bonds with Carbon”).
- Identify the important biological functional groups (see Figure 4.9).
- Understand the chemical nature of functional groups (see Figure 4.9).

The following questions provide a quick check of your basic knowledge in these areas.

Figure showing carbon B with a single line connected to another C and a double line connected to a different C.

1. How many hydrogen atoms can be attached to carbon B?

That's correct.

Correct! If you had trouble with this question, review the following material: Each carbon atom can form four bonds. A double bond counts as two bonds, so carbon B has a total of three bonds so far, and it can form one more single bond with a hydrogen atom.

2

That's incorrect. Try again.

Incorrect. Review the following material: Each carbon atom can form four bonds. A double bond counts as two bonds, so carbon B has a total of three bonds so far, and it can form one more single bond with a hydrogen atom.

3

That's incorrect. Try again.

Incorrect. Review the following material: Each carbon atom can form four bonds. A double bond counts as two bonds, so carbon B has a total of three bonds so far, and it can form one more single bond with a hydrogen atom.

4

That's incorrect. Try again.

Incorrect. Review the following material: Each carbon atom can form four bonds. A double bond counts as two bonds, so carbon B has a total of three bonds so far, and it can form one more single bond with a hydrogen atom.

A. -OH, B. C with a double bond to O and a single bond to OH, C. -NH2, D. -SH

2. Which functional group(s) is (are) present in all amino acids?

```
B and D
```

That's incorrect. Try again.

Incorrect. Review the following material: An amino acid has an amino group (-NH2) and a carboxyl group (-COOH).

```
A and B
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That's incorrect. Try again.

Incorrect. Review the following material: An amino acid has an amino group (-NH2) and a carboxyl group (-COOH).

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B and C
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That's correct.

Correct! If you had trouble with this question, review the following material: An amino acid has an amino group (-NH2) and a carboxyl group (-COOH).

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C only
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That's incorrect. Try again.

Incorrect. Review the following material: An amino acid has an amino group (-NH2) and a carboxyl group (-COOH).

3. Which functional group can act as an acid?

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amino
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That's incorrect. Try again.

Incorrect. Review the following material: An acid can donate H+. The carboxyl group (-COOH) can donate H+ (forming -COO-) because the covalent bond between oxygen and hydrogen is so polar. It is the carboxyl group of amino acids that makes them acids.

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carbonyl
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That's incorrect. Try again.

Incorrect. Review the following material: An acid can donate H+. The carboxyl group (-COOH) can donate H+ (forming -COO-) because the covalent bond between oxygen and hydrogen is so polar. It is the carboxyl group of amino acids that makes them acids.

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methyl
```

That's incorrect. Try again.

Incorrect. Review the following material: An acid can donate H+. The carboxyl group (-COOH) can donate H+ (forming -COO-) because the covalent bond between oxygen and hydrogen is so polar. It is the carboxyl group of amino acids that makes them acids.

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carboxyl
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That's correct.

Correct! If you had trouble with this question, review the following material: An acid can donate H+. The carboxyl group (-COOH) can donate H+ (forming -COO-) because the covalent bond between oxygen and hydrogen is so polar. It is the carboxyl group of amino acids that makes them acids.

4. Which functional group can act as a base?

```
amino
```

That's correct.

Correct! If you had trouble with this question, review the following material: A base can accept H+ from the surrounding solution. The amino group (-NH2) can accept H+ from the surrounding solution, forming -NH3+. Therefore, the amino group can act as a base.

```
carbonyl
```

That's incorrect. Try again.

Incorrect. Review the following material: A base can accept H+ from the surrounding solution. The amino group (-NH2) can accept H+ from the surrounding solution, forming -NH3+. Therefore, the amino group can act as a base.

```
carboxyl
```

That's incorrect. Try again.

Incorrect. Review the following material: A base can accept H+ from the surrounding solution. The amino group (-NH2) can accept H+ from the surrounding solution, forming -NH3+. Therefore, the amino group can act as a base.

```
methyl
```

That's incorrect. Try again.

Incorrect. Review the following material: A base can accept H+ from the surrounding solution. The amino group (-NH2) can accept H+ from the surrounding solution, forming -NH3+. Therefore, the amino group can act as a base.

Learning Objectives

After studying this chapter, you should be able to do the following:

- 5.1Explain the processes by which polymers are assembled and disassembled.
- 5.2Describe the structures and functions of simple and complex carbohydrates.
- 5.3Describe the structures of three kinds of lipids and explain their functions.
- 5.4List the main functions of proteins and describe the structures of amino acids and proteins.
- 5.5Compare and contrast the structures of DNA and RNA and their component nucleotides, and describe the functions of these polynucleotides.
- 5.6Explain how our ability to rapidly sequence DNA has affected biological inquiry and applications.

Concept 5.1: Macromolecules are polymers, built from monomers

Large carbohydrates, proteins, and nucleic acids, also known as **macromolecules** for their huge size, are chain-like molecules called polymers (from the Greek *polys*, many, and *meros*, part). A **polymer** is a long molecule consisting of many similar or identical building blocks linked by covalent bonds, much as a train consists of a chain of boxcars. The repeating units that serve as the building blocks of a polymer are smaller molecules called **monomers** (from the Greek *monos*, single). In addition to forming polymers, some monomers have functions of their own.

The Synthesis and Breakdown of Polymers

Although each class of polymer is made up of a different type of monomer, the chemical mechanisms by which cells make polymers (polymerization) and break them down are similar for all classes of large biological molecules. In cells, these processes are facilitated by **enzymes**, specialized macromolecules (usually proteins) that speed up chemical reactions. The reaction that connects a monomer to another monomer or a polymer is a *condensation reaction*, a reaction in which two molecules are covalently bonded to each other with the loss of a small molecule. If a water molecule is lost, it is known as a **dehydration reaction**. For example, carbohydrate and protein polymers are synthesized by dehydration reactions. Each reactant contributes part of the water molecule that is released during the reaction: One provides a hydroxyl group —OH, while the other provides a hydrogen —H **(Figure 5.2a)**. This reaction is repeated as monomers are added to the chain one by one, lengthening the polymer.

Figure 5.2

The synthesis and breakdown of carbohydrate and protein polymers.

The synthesis and breakdown of carbohydrate and protein polymers.

Part a. Dehydration reaction: synthesizing a polymer. The loss of a water molecule results in a covalent bond that joins two shorter polymers into a longer polymer. Two short polymers represented by linked circles (monomers) each have an O H on one end and an H on the opposite end. Dehydration removes a water molecule (an H from the end of one short polymer combines with the O H from the end of the other short polymer, forming H 2 O). With removal of water, a covalent bond forms between the two shorter polymers, forming one longer polymer.

Part b. Hydrolysis: breaking down a polymer. A water molecule is added to a longer polymer, breaking a bond, which results in two shorter polymers. Hydrolysis adds a water molecule, breaking a bond in a longer polymer. The result is two shorter polymers, one with an H on the end, and the other with an O H group on the end (both from the water molecule).

Animation

Making and Breaking Polymers

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Polymers are disassembled to monomers by **hydrolysis**, a process that is essentially the reverse of the dehydration reaction **(Figure 5.2b)**. Hydrolysis means water breakage (from the Greek *hydro*, water, and *lysis*, break). The bond between monomers is broken by the addition of a water molecule, with a hydrogen from water attaching to one monomer and the hydroxyl group attaching to the other. An example of hydrolysis within our bodies is the process of digestion. The bulk of the organic material in our food is in the form of polymers that are much too large to enter our cells. Within the digestive tract, various enzymes attack the polymers, speeding up hydrolysis. Released monomers are then absorbed into the bloodstream for distribution to all body cells. Those cells can then use dehydration reactions to assemble the monomers into new, different polymers that can perform specific functions required by the cell. (Dehydration reactions and hydrolysis can also be involved in the formation and breakdown of molecules that are not polymers, such as some lipids.)

The Diversity of Polymers

A cell has thousands of different macromolecules; the collection varies from one type of cell to another. The inherited differences between close relatives, such as human siblings, reflect small variations in polymers, particularly DNA and proteins. Molecular differences between unrelated individuals are more extensive, and those between species greater still. The diversity of macromolecules in the living world is vast, and the possible variety is effectively limitless.

What is the basis for such diversity in life's polymers? These molecules are constructed from only 40 to 50 common monomers and some others that occur rarely. Building a huge variety of polymers from such a limited number of monomers is analogous to constructing hundreds of thousands of words from only 26 letters of the alphabet. The key is arrangement—the particular linear sequence that the units follow. However, this analogy falls far short of describing the great diversity of macromolecules because most biological polymers have many more monomers than the number of letters in even the longest word. Proteins, for example, are built from 20 kinds of amino acids arranged in chains that are typically hundreds of amino acids long. The molecular logic of life is simple but elegant: Small molecules common to all organisms act as building blocks that are ordered into unique macromolecules.

Despite this immense diversity, molecular structure and function can still be grouped roughly by class. Let's examine each of the four major classes of large biological molecules. For each class, the large molecules have emergent properties not found in their individual components.

Concept Check 5.1

- What are the four main classes of large biological molecules? Which class does not consist of polymers?
- How many molecules of water are needed to completely hydrolyze a polymer that is ten monomers long?
- **WHAT IF?** If you eat a piece of fish, what reactions must occur for the amino acid monomers in the protein of the fish to be converted to new proteins in your body?

*For suggested answers, see*Appendix A.**

Concept 5.2: Carbohydrates serve as fuel and building material

Carbohydrates include sugars and polymers of sugars. The simplest carbohydrates are the monosaccharides, or simple sugars; these are the monomers from which more complex carbohydrates are built. Disaccharides are double sugars, consisting of two monosaccharides joined by a covalent bond. Carbohydrate macromolecules are polymers called polysaccharides, composed of many sugar building blocks.

Animation

Carbohydrates

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Sugars

Monosaccharides (from the Greek *monos*, single, and *sacchar*, sugar) generally have molecular formulas that are some multiple of the unit CH2O. Glucose C6H12O6, the most common monosaccharide, is of central importance in the chemistry of life. In the structure of glucose, we can see the trademarks of a monosaccharide: The molecule has a carbonyl group, ╲╱C═O, and multiple hydroxyl groups, —OH** (**Figure 5.3**)**. Depending on the location of the carbonyl group, a monosaccharide is either an aldose (aldehyde sugar) or a ketose (ketone sugar). Glucose, for example, is an aldose; fructose, an isomer of

glucose, is a ketose. (Most names for sugars end in -*ose*.) Another criterion for classifying monosaccharides is the size of the carbon skeleton, which ranges from three to seven carbons long. Glucose, fructose, and other sugars that have six carbons are called hexoses. Trioses (three-carbon sugars) and pentoses (five-carbon sugars) are also common.

Figure 5.3

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The structure and classification of some monosaccharides.
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Structure and classification of some monosaccharides.

Sugars vary in the location of their carbonyl groups (orange), the length of their carbon skeletons, and the way their parts are arranged spatially around asymmetric carbons (compare, for example, the purple portions of glucose and galactose).

Sugars can be classified by the number of carbons, the location of the carbonyl group, and the spatial arrangement at each asymmetric carbon.

The number of carbons in a sugar molecule is one way to classify it. For example, a triose is a sugar with three carbons and the formula C 3 H 6 O 3. Examples include glyceraldehyde and dihydroxyacetone, which are both initial breakdown products of glucose. A pentose has five carbons and the formula C 5 H 10 O 5. Examples of pentoses are ribose, a component in RNA and ribulose, an intermediate in photosynthesis. A hexose has six carbons and the formula C 6 H 12 O 6. Glucose, galactose and fructose are hexoses used as energy sources for organisms.

The location of the carbonyl group (C double bonded to O) on the carbon chain is another way to further classify a sugar. A sugar that has its carbonyl group at the end of the chain (an aldehyde) is called an aldose. Glyceraldehyde, ribose, glucose and galactose are aldoses. When the carbonyl group is not on the end of the carbon chain, it is a ketone and the sugar is called a ketose. Dihydroxyacetone, ribulose and fructose are ketoses.

The position of the hydroxyl groups (O H) on each asymmetric carbon is the most specific way to classify sugars. For example, glucose and galactose both have six carbons. Drawn vertically with a linear carbon skeleton (Fischer projection), the first (top) carbon is part of an aldehyde (C H O). The remaining carbons have O H groups on either the left or the right side of the chain, and a hydrogen atom across from each O H. The only difference between the two sugars is the location of the O H on the fourth carbon. This O H is on the right in glucose, but on the left in galactose. In all other ways, both sugars are the same. The second and fifth carbons have an OH on the right. The third carbon has O H on the left. The sixth carbon is not asymmetric, and has two hydrogens and an O H. Fructose has the same structure as glucose, but, since it is a

ketose, it has only a carbonyl group on carbon 2. The first carbon in fructose is a C H 2 O H, which is not asymmetric.

MAKE CONNECTIONS *In the 1970s, a process was developed that converts the glucose in corn syrup to its sweeter-tasting isomer, fructose. High-fructose corn syrup, a common ingredient in soft drinks and processed food, is a mixture of glucose and fructose. What type of isomers are glucose and fructose? (See*Figure 4.7.)*

*For suggested answer, see*Appendix A.*

Animation

Monosaccharides

image

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Still another source of diversity for simple sugars is in the way their parts are arranged spatially around asymmetric carbons. (Recall that an asymmetric carbon is a carbon attached to four different atoms or groups of atoms.) Glucose and galactose, for example, differ only in the placement of parts around one asymmetric carbon (see the purple boxes in **Figure 5.3**). What seems like a small difference is significant enough to give the two sugars distinctive shapes and binding activities, thus different behaviors.

Although it is convenient to draw glucose with a linear carbon skeleton, this representation is not completely accurate. In aqueous solutions, glucose molecules, as well as most other five- and six-carbon sugars, form rings, because they are the most stable form of these sugars under physiological conditions **(Figure 5.4)**.

Figure 5.4

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Linear and ring forms of glucose.
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The illustration gives the linear and ring forms of glucose.

Chemical equilibrium between the linear and ring structures greatly favors the formation of rings. The carbons of the sugar are numbered 1 to 6. To form the glucose ring, carbon 1 bonds to the oxygen attached to carbon 5. The linear form is a vertical chain where carbons are numbered from top to bottom. C 1 is double bonded to O up and right, and single bonded to H up and left. Carbons 2 through 6 are single bonded to H

and single bonded to O H, and C 6 is also single bonded to another H. When a ring forms, the O from O H on C 5 bonds to C 1. The H from that O H then moves to the double bonded O from C 1. This forms a single bond to O H from C 1. The ring has 5 C atoms and 1 O atom. C 5 is single bonded to C H 2 O H, where that C is C 6.

DRAW IT *Start with the linear form of fructose (see*Figure 5.3*)* and draw the formation of the fructose ring in two steps, as shown in (a). First, number the carbons starting at the top of the linear structure. Then draw the molecule in a ringlike orientation, attaching carbon*5*via its oxygen to carbon*2*. Compare the number of carbons in the ring portions of fructose and glucose.*

*For suggested answer, see**Appendix A*.

Monosaccharides, particularly glucose, are major nutrients for cells. In the process known as cellular respiration, cells extract energy from glucose molecules by breaking them down in a series of reactions. Not only are monosaccharides a major fuel for cellular work, but their carbon skeletons also serve as raw material for the synthesis of other types of small organic molecules, such as amino acids and fatty acids. Monosaccharides that are not immediately used in these ways are generally incorporated as monomers into disaccharides or polysaccharides, discussed next.

A **disaccharide** consists of two monosaccharides joined by a **glycosidic linkage**, a covalent bond formed between two monosaccharides by a dehydration reaction (*glyco* refers to carbohydrate). For example, maltose is a disaccharide formed by the linking of two molecules of glucose **(Figure 5.5a)**. Also known as malt sugar, maltose is an ingredient used in brewing beer. The most prevalent disaccharide is sucrose, or table sugar. Its two monomers are glucose and fructose **(Figure 5.5b)**. Plants generally transport carbohydrates from leaves to roots and other nonphotosynthetic organs in the form of sucrose. Lactose, the sugar present in milk, is another disaccharide, in this case a glucose molecule joined to a galactose molecule. Disaccharides must be broken down into monosaccharides to be used for energy by organisms. Lactose intolerance is a common condition in humans who lack lactase, the enzyme that breaks down lactose. The sugar is instead broken down by intestinal bacteria, causing formation of gas and subsequent cramping. The problem may be avoided by taking the enzyme lactase when eating or drinking dairy products or consuming dairy products that have already been treated with lactase to break down the lactose.

Figure 5.5

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Examples of disaccharide synthesis.
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Abbreviated ring structure and Dehydration reaction in the synthesis of (a) maltose and (b)sucrose.

Abbreviated ring structure. Each unlabeled corner represents a carbon. The ring's thicker edge indicates that you are looking at the ring edge-on; the components attached to the ring lie above or below the plane of the ring. The ring form of glucose is drawn using an edge-on view of a hexagon. The points of the hexagon are on

the right and left, and the carbons are numbered starting with the carbon on the far right, which was the aldehyde carbon in the linear form of the sugar. Each carbon is bonded to an O H and a hydrogen, one above the ring, and the other below it. Carbons 2 and 3 are on the near side of the hexagon, carbon 4 is the left point, and carbon 5 is on the far side. The oxygen of carbon 5 completes the hexagon, as it is bonded to carbon 1. Carbon 6, which is a C H 2 O H is bonded to carbon 5 and drawn above the ring. In glucose, the O H groups are below the ring on carbons 2 and 4, and above the ring on carbon 3. The O H on carbon 1 can be either below or above the ring, depending on whether the ring is in the alpha or beta form, respectively (see Figure 5.7a).

(a)Dehydration reaction in the synthesis of maltose. The bonding of two glucose units forms maltose. The 1-4 glycosidic linkage joins the number 1 carbon of one glucose to the number 4 carbon of the second glucose. Joining the glucose monomers in a different way would result in a different disaccharide. Two glucose molecules can form a bond to make a disaccharide. In the case of maltose, a 1-4 glycosidic linkage is formed between two glucose molecules. This means that carbon 1 of the first glucose molecule is bonded to the oxygen on carbon 4 for the second glucose molecule. When this bond forms, the O H on carbon 1 of the first glucose and the H from the hydroxyl group on the second glucose leave and make a water molecule. This dehydration reaction leads to the formation of a disaccharide. If two glucose molecules bonded in a different pattern, a different disaccharide would result.

(b)Dehydration reaction in the synthesis of sucrose. Sucrose is a disaccharide formed from glucose and fructose. Notice that fructose forms a five-sided ring, though it is a hexose-like glucose. The structure of fructose is as follows: Each unlabeled corner represents a carbon. The ring's thicker edge indicates that you are looking at the ring edge-on; the components attached to the ring lie above or below the plane of the ring. The ring form of fructose is drawn using an edge-on view of a pentagon. The points of the pentagon are on the right and left, and the carbons are numbered starting with the carbon on the far right. Each carbon is bonded to an O H and hydrogen, one above the ring, and the other below it. Carbons 3 and 4 are on the near side of the hexagon, carbon 5 is the left point, and carbon 6 is on the far side. Carbon 1 is part of C H 2 O H bonded to carbon 2 of the ring. The oxygen of carbon 5 completes the pentagon, as it is bonded to carbon 1. Carbon 6, which is a C H 2 O H is bonded to carbon 5 and drawn above the ring. In fructose, the O H groups are below the ring on carbons 3 and 4. The bonding of one glucose unit and one fructose unit forms sucrose. The 1-2 glycosidic linkage joins the number 1 carbon of glucose to the number 2 carbon of fructose. Joining the monomers in a different way would result in a different disaccharide. When this bond forms, the O H on carbon 1 of the glucose unit and the H from the hydroxyl group on the fructose unit leave and make a water molecule. This dehydration reaction leads to the formation of a disaccharide.

DRAW IT *Referring to**Figures 5.3**and*5.4*, number the carbons in each sugar in this figure. How does the name of each linkage relate to the numbers?*

*For suggested answer, see**Appendix A*.

Animation

Synthesis of Sucrose

image

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image

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Polysaccharides

Polysaccharides are macromolecules, polymers with a few hundred to a few thousand monosaccharides joined by glycosidic linkages. Some polysaccharides serve as storage material, hydrolyzed as needed to provide monosaccharides for cells. Other polysaccharides serve as building material for structures that protect the cell or the whole organism. The architecture and function of a polysaccharide are determined by its monosaccharides and by the positions of its glycosidic linkages.

Storage Polysaccharides

Both plants and animals store sugars for later use in the form of storage polysaccharides **(Figure 5.6)**. Plants store **starch**, a polymer of glucose monomers, as granules within cellular structures known as plastids. (Plastids include chloroplasts.) Synthesizing starch enables the plant to stockpile surplus glucose. Because glucose is a major cellular fuel, starch represents stored energy. The sugar can later be withdrawn by the plant from this carbohydrate “bank” by hydrolysis, which breaks the bonds between the glucose monomers. Most animals, including humans, also have enzymes that can hydrolyze plant starch, making glucose available as a nutrient for cells. Potato tubers and grains—the fruits of wheat, maize (corn), rice, and other grasses—are the major sources of starch in the human diet.

Figure 5.6

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    Polysaccharides of plants and animals.
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Gardener holding potatoes; arrows flow to microscopic image depicting storage structures called plastids containing starch granules in potato tuber cell. Diagram of starch structure flows from image.

Two starch structures depicted: amylose starch and amylopectin.

Amylose starch consists of a long, unbranched chain of glucose (6 carbon ring) monomers joined by 1-4 linkages (number 1 carbon to number 4 carbon).

Amylopectin starch consists of a somewhat branched chain of glucose monomers with 1-6 linkages at the branch points. One branch is depicted in the diagram.

Arrow flows from gardener's arm to microscopic image of glycogen granules stored in muscle tissues (scale: 50 micrometers). Diagram of glycogen structure flows from image. Glycogen is a branched polysaccharide composed of glucose monomers. Several branches depicted in the diagram, indicating that glycogen is more extensively branched than amylopectin starch. Arrow flows from leaf to microscopic image of plant cell with cell wall visible; another arrow flows from cell wall to image of cellulose. Structure of cellulose flows from image.

(a) Starch stored in plant cells, **(b)** glycogen stored in muscle cells, and **(c)** structural cellulose fibers in plant cell walls are all polysaccharides composed entirely of glucose monomers (green hexagons). In starch and glycogen, the polymer chains tend to form helices in unbranched regions because of the angle of the 1–4 linkages between glucose molecules. There are two kinds of starch: amylose and amylopectin. Cellulose, with a different kind of glucose linkage, is always unbranched.

Most of the glucose monomers in starch are joined by 1–4 linkages (number 1 carbon to number 4 carbon), like the glucose units in maltose (see Figure 5.5a). The simplest form of starch, amylose, is unbranched. Amylopectin, a more complex starch, is a branched polymer with 1–6 linkages at the branch points. Both of these starches are shown in **Figure 5.6a**.

Animals store a polysaccharide called **glycogen**, a polymer of glucose that is like amylopectin but more extensively branched **(Figure 5.6b)**. Vertebrates store glycogen mainly in liver and muscle cells. Breakdown of glycogen in these cells releases glucose when the demand for energy increases. (The extensively branched structure of glycogen fits its function: More free ends are available for breakdown.) This stored fuel cannot sustain an animal for long, however. In humans, for example, glycogen stores are depleted in about a day unless they are replenished by eating. This is an issue of concern in low-carbohydrate diets, which can result in weakness and fatigue.

Structural Polysaccharides

Organisms build strong materials from structural polysaccharides. For example, the polysaccharide called **cellulose** is a major component of the tough walls that enclose plant cells **(Figure 5.6c)**. Globally, plants produce almost 1014 kg 100 billion tons of cellulose per year; it is the most abundant organic compound on Earth.

Like starch, cellulose is a polymer of glucose with 1–4 glycosidic linkages, but the linkages in these two polymers differ. The difference is based on the fact that there are actually two slightly different ring

structures for glucose **(Figure 5.7a)**. When glucose forms a ring, the hydroxyl group attached to the number 1 carbon is positioned either below or above the plane of the ring. These two ring forms for glucose are called alpha α and beta β, respectively. (Greek letters are often used as a "numbering" system for different versions of biological structures, much as we use the letters a, b, c, and so on for the parts of a question or a figure.) In starch, all the glucose monomers are in the α configuration **(Figure 5.7b)**, the arrangement we saw in Figures 5.4 and 5.5. In contrast, the glucose monomers of cellulose are all in the β configuration, making every glucose monomer "upside down" with respect to its neighbors (**Figure 5.7c**; see also Figure 5.6c).

Figure 5.7

Starch and cellulose structures.

Starch and cellulose structures.

Part a. Alpha and beta glucose ring structures. These two interconvertible forms of glucose differ in the placement of the hydroxyl group attached to the number 1 carbon. Whenever a sugar forms a ring in solution, there are two possible structures, depending on how the ring closes. For example, in glucose, if the ring closes with the OH on carbon 1 oriented down, then the sugar is in the alpha form. If the ring closes with the OH on this carbon pointed up, then the sugar is in the beta form. The two forms can open up and close again in solution, so they are in chemical equilibrium with each other and the linear form of the sugar.

Part b. Starch: 1-4 linkage of alpha glucose monomers. All monomers are in the same orientation. Starch is a polysaccharide of alpha glucose. Starch is the result of assembling alpha glucose into a polysaccharide using 1-4 glycosidic bonds. All monomers are in the same orientation. The resulting starch molecule has an overall helical shape.

Part c. Cellulose is a polysaccharide of beta glucose with 1-4 glycosidic linkages. Every beta glucose monomer is upside down with respect to its neighbors. Cellulose is the result of assembling beta glucose into a polysaccharide using 1-4 glycosidic bonds. Every beta glucose monomer is upside down with respect to its neighbors. The resulting cellulose molecule is straight and rather rigid.

Animation

Starch, Cellulose, and Glycogen Structures

image

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The differing glycosidic linkages in starch and cellulose give the two molecules distinct three-dimensional shapes. Certain starch molecules are largely helical, fitting their function of efficiently storing glucose units. Conversely, a cellulose molecule is straight. Cellulose is never branched, and some hydroxyl groups on its glucose monomers are free to hydrogen-bond with the hydroxyls of other cellulose molecules lying parallel to it. In plant cell walls, parallel cellulose molecules held together in this way are grouped into units called microfibrils (see Figure 5.6c). These cable-like microfibrils are a strong building material for plants and an important substance for humans because cellulose is the major constituent of paper and the only component of cotton. The unbranched structure of cellulose thus fits its function: imparting strength to parts of the plant.

Enzymes that digest starch by hydrolyzing its α linkages are unable to hydrolyze the β linkages of cellulose due to the different shapes of these two molecules. In fact, few organisms possess enzymes that can digest cellulose. Almost all animals, including humans, do not; the cellulose in our food passes through the digestive tract and is eliminated with the feces. Along the way, the cellulose abrades the wall of the digestive tract and stimulates the lining to secrete mucus, which aids in the smooth passage of food through the tract. Thus, although cellulose is not a nutrient for humans, it is an important part of a healthy diet. Most fruits, vegetables, and whole grains are rich in cellulose. On food packages, “insoluble fiber” refers mainly to cellulose (see Figure 5.7).

Some microorganisms can digest cellulose, breaking it down into glucose monomers. A cow harbors cellulose-digesting prokaryotes and protists in its gut. These microbes hydrolyze the cellulose of hay and grass and convert the glucose to other compounds that nourish the cow. Similarly, a termite, which is unable to digest cellulose by itself, has prokaryotes or protists living in its gut that can make a meal of wood. Some fungi can also digest cellulose in soil and elsewhere, thereby helping recycle chemical elements within Earth’s ecosystems.

Another important structural polysaccharide is **chitin**, the carbohydrate used by arthropods (insects, spiders, crustaceans, and related animals) to build their exoskeletons—hard cases that surround the soft parts of an animal **(Figure 5.8)**. Made up of chitin embedded in a layer of proteins, the case is leathery and flexible at first, but becomes hardened when the proteins are chemically linked to each other (as in insects) or encrusted with calcium carbonate (as in crabs). Chitin is also found in fungi, which use this polysaccharide rather than cellulose as the building material for their cell walls. Chitin is similar to cellulose, with β linkages, except that the glucose monomer of chitin has a nitrogen-containing attachment (see Figure 5.8).

Figure 5.8

Chitin, a structural polysaccharide.

Chitin, a structural polysaccharide. Part a. The chitin monomer structure. Part b. Chitin in nature.

Part a. The chitin monomer is N-acteylglucosamine, a derivative of glucose where the alcohol on carbon 2 is replaced with an acetamido group.

Part b. Chitin, embedded in proteins, forms the exoskeleton of arthropods. This picture shows a cicada molting, shedding its old exoskeleton and emerging in adult form.

Concept Check 5.2

- Write the formula for a monosaccharide that has three carbons.
- A dehydration reaction joins two glucose molecules to form maltose. The formula for glucose is $C_6H_{12}O_6$. What is the formula for maltose?
- **WHAT IF?** After a cow is given antibiotics to treat an infection, a vet gives the animal a drink of "gut culture" containing various prokaryotes. Why is this necessary?

*For suggested answers, see*Appendix A.**

Concept 5.3: Lipids are a diverse group of hydrophobic molecules

Lipids are the one class of large biological molecules that does not include true polymers, and they are generally not big enough to be considered macromolecules. The compounds called **lipids** are grouped with each other because they share one important trait: They are hydrophobic: They mix poorly, if at all, with water. This behavior of lipids is based on their molecular structure. Although they may have some polar bonds associated with oxygen, lipids consist mostly of hydrocarbon regions with relatively nonpolar C—H bonds. Lipids are varied in form and function. They include waxes and certain pigments, but we will focus on the types of lipids that are most important biologically: fats, phospholipids, and steroids.

Animation

Lipids

image

Play

image

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Fats

Although fats are not polymers, they are large molecules assembled from smaller molecules by dehydration reactions, like the dehydration reaction described in Figure 5.2a. A **fat** consists of a glycerol molecule joined to three fatty acids **(Figure 5.9)**. Glycerol is an alcohol; each of its three carbons bears a hydroxyl group. A **fatty acid** has a long carbon skeleton, usually 16 or 18 carbon atoms in length. The carbon at one end of the skeleton is part of a carboxyl group, the functional group that gives these molecules the name fatty *acid*. The rest of the skeleton consists of a hydrocarbon chain. The relatively nonpolar C—H bonds in the hydrocarbon chains of fatty acids are the reason fats are hydrophobic. Fats separate from water because the water molecules hydrogen-bond to one another and exclude the fats. This is why vegetable oil (a liquid fat) separates from the aqueous vinegar solution in a bottle of salad dressing.

Figure 5.9

The synthesis and structure of a fat, or triacylglycerol.

The synthesis and structure of a fat, or triacylglycerol.

Part a. One of three dehydration reactions in the synthesis of a fat: synthesis of a triacylglycerol molecule. The formation of a triacylglycerol, or fat molecule, begins when a molecule of glycerol reacts with a fatty acid in a dehydration reaction. Glycerol is a three carbon alcohol with one O H group on each carbon. A fatty acid is a long carbon chain with a carboxyl (C O 2 H) group at one end. The O H of the glycerol loses its hydrogen when it bonds to the carbon of the carboxyl group. The hydrogen from the glycerol and the OH from the carboxyl group leave and form a water molecule, and the glycerol and fatty acid are joined in an ester linkage.

Part b. Structure of a triacylglycerol (fat) molecule. A triacylglycerol is formed when a dehydration reaction occurs between each of the three OH groups on the glycerol molecule and a total of three fatty acid molecules. These fatty acids may be the same or different from one another. The resulting molecule of fat is held together by three ester linkages.

The molecular building blocks of a fat are one molecule of glycerol and three molecules of fatty acids. The carbons of the fatty acids are arranged zigzag to suggest the actual orientations of the four single bonds extending from each carbon (see Figures 4.3a and 4.6b).

In making a fat, each fatty acid molecule is joined to glycerol by a dehydration reaction **(Figure 5.9a)**. This results in an ester linkage, a bond between a hydroxyl group and a carboxyl group. The completed fat consists of three fatty acids linked to one glycerol molecule. (Other names for a fat are ***triacylglycerol*** and

triglyceride; levels of triglycerides are reported when blood is tested for lipids.) The fatty acids in a fat can all be the same, or they can be of two or three different kinds, as in **Figure 5.9b**.

The terms *saturated* fats and *unsaturated* fats are commonly used in the context of nutrition **(Figure 5.10)**. These terms refer to the structure of the hydrocarbon chains of the fatty acids. If there are no double bonds between carbon atoms composing a chain, then as many hydrogen atoms as possible are bonded to the carbon skeleton. Such a structure is said to be *saturated* with hydrogen, and the resulting fatty acid is therefore called a **saturated fatty acid(Figure 5.10a)**. An **unsaturated fatty acid** has one or more double bonds, with one fewer hydrogen atom on each double-bonded carbon. Nearly every double bond in naturally occurring fatty acids is a *cis* double bond, which creates a kink in the hydrocarbon chain wherever it occurs **(Figure 5.10b)**. (See Figure 4.7b to remind yourself about *cis* and *trans* double bonds.)

Figure 5.10

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Saturated and unsaturated fats and fatty acids.
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Saturated and unsaturated fats and fatty acids.

Part a. At room temperature, the molecules of a saturated fat, such as the fat in butter, are packed closely together, forming a solid. In the structural formula of a saturated fat molecule, the fat is a glycerol joined to three fatty acid “tails” through ester linkages. Each hydrocarbon chain (tail) is represented as a zigzag line, where each bend represents a carbon atom and hydrogen atoms are not shown. The space-filling model of stearic acid, a saturated fatty acid, shows the carbon in the ester linkage bonded to one oxygen atom, double bonded to another oxygen atom, and then bonded to an unkinked (straight) chain of carbon and hydrogen atoms in the “tail” of the saturated fatty acid.

Part b. At room temperature, the molecules of an unsaturated fat such as olive oil cannot pack together closely enough to solidify because of the kinds in some of their fatty acid hydrocarbon chains. The structural formula of an unsaturated fat molecule (glycerol joined to three fatty acid tails) shows a double bond in one of the hydrocarbon chains. A space-filling model of oleic acid, an unsaturated fatty acid, shows that the double bond in the hydrocarbon chain depicted in the structural formula is a Cis double bond, which results in bending (a kink).

A fat made from saturated fatty acids is called a saturated fat. Most animal fats are saturated: The hydrocarbon chains of their fatty acids—the “tails” of the fat molecules—lack double bonds (see Figure 5.10a), and their flexibility allows the fat molecules to pack together tightly. Saturated animal fats, such as lard and butter, are solid at room temperature. In contrast, the fats of plants and fishes are generally unsaturated, meaning that they are composed of one or more types of unsaturated fatty acids. Usually liquid at room temperature, plant and fish fats are referred to as oils—olive oil and cod liver oil are examples. The

kinks where the *cis* double bonds are located (see Figure 5.10b) prevent the molecules from packing together closely enough to solidify at room temperature. The phrase *hydrogenated vegetable oils* on food labels means that unsaturated fats have been synthetically converted to saturated fats by adding hydrogen, allowing them to solidify. Peanut butter, margarine, and many other products are hydrogenated to prevent lipids from separating out in liquid (oil) form.

A diet rich in saturated fats is one of several factors that may contribute to the cardiovascular disease known as atherosclerosis. In this condition, deposits called plaques develop within the walls of blood vessels, causing inward bulges that impede blood flow and reduce the resilience of the vessels. The process of hydrogenating vegetable oils produces not only saturated fats but also unsaturated fats with *trans* double bonds. It appears that ***trans*fats** can contribute to coronary heart disease (see Concept 42.4). Because *trans* fats have been especially common in baked goods and processed foods, the U.S. Food and Drug Administration (FDA) requires nutritional labels to include information on *trans* fat content. In addition, the FDA has ordered U.S. food manufacturers to stop adding *trans* fats to foods. Globally, the World Health Organization has a stated goal of eliminating industrially produced *trans* fats by 2023; over a third of countries worldwide have already passed such regulations.

The major function of fats is energy storage. The hydrocarbon chains of fats are similar to gasoline molecules and just as rich in energy. A gram of fat stores more than twice as much energy as a gram of a polysaccharide, such as starch. Because plants are relatively immobile, they can function with bulky energy storage in the form of starch. (Vegetable oils are generally obtained from seeds, where more compact storage is an asset to the plant.) Animals, however, must carry their energy stores with them, so there is an advantage to having a more compact reservoir of fuel—fat. Humans and other mammals stock their long-term food reserves in adipose cells (see Figure 4.6a), which swell and shrink as fat is deposited and withdrawn from storage. In addition to storing energy, adipose tissue also cushions such vital organs as the kidneys, and a layer of fat beneath the skin insulates the body. This subcutaneous layer is especially thick in whales, seals, and most other marine mammals, insulating their bodies in cold ocean water.

Phospholipids

Cells as we know them could not exist without another type of lipid, called phospholipids. Phospholipids are essential for cells because they are major constituents of cell membranes. Their structure provides a classic example of how form fits function at the molecular level. As shown in **Figure 5.11**, a **phospholipid** is similar to a fat molecule but has only two fatty acids attached to glycerol rather than three. The third hydroxyl group of glycerol is joined to a phosphate group, which has a negative electrical charge in the cell. Typically, an additional small charged or polar molecule is also linked to the phosphate group. Choline is one such molecule (see Figure 5.11), but there are many others as well, allowing formation of a variety of phospholipids that differ from each other.

Figure 5.11

The structure of a phospholipid.

The structure of a phospholipid.

Part a. Structural formula of a phospholipid. A charged molecule is linked to a negatively charged phosphate group, which is linked to a glycerol attached to two hydrophobic fatty acid tails. The phospholipid consists of a hydrophilic (charged) head and two hydrophobic tails. Choline (a nitrogen-containing molecule with a positive charge on N) is linked to a phosphate group (a central P double bonded to one O and single bonded to three other oxygen atoms with a negative charge on one oxygen atom). The phosphate group is joined to the third hydroxyl group of glycerol, which is attached to two fatty acid tails (hydrophobic chains of carbon and hydrogen; in one tail there is a kink due to a cis double bond).

Part b. Space filling model of a phospholipid. Choline molecule contains N and is linked to the phosphate group containing P, which is linked to a glycerol attached to two fatty acids.

Part c. Phospholipid symbol: An oval represents the hydrophilic head; is attached to two squiggly lines representing the hydrophobic tails.

Part d. Phospholipid bilayer drawn with phospholipid symbols. Heads contact water on both sides of the membrane, tails point at each other inside the membrane.

A phospholipid has a hydrophilic (polar) head and two hydrophobic (nonpolar) tails. This particular phospholipid, called a phosphatidylcholine, has a choline attached to a phosphate group. Shown here are **(a)** the structural formula, **(b)** the space-filling model (yellow = phosphorus, blue = nitrogen), **(c)** the symbol for a phospholipid that will appear throughout this book, and **(d)** the bilayer structure formed by selfassembly of phospholipids in an aqueous environment.

DRAW IT *Draw an oval around the hydrophilic head of the space-filling model.*

*For suggested answer, see**Appendix A.*

Figure 5-11 Walkthrough

The structure of a phospholipid

Watch an explanation of this figure in the Figure 5.11 Walkthrough: The structure of a phospholipid.

image

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Animation

Space-Filling Model of a Phospholipid

image

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The two ends of phospholipids show different behaviors with respect to water. The hydrocarbon tails are hydrophobic and are excluded from water. However, the phosphate group and its attachments form a hydrophilic head that has an affinity for water. When phospholipids are added to water, they self-assemble into a double-layered sheet called a “bilayer” that shields their hydrophobic fatty acid tails from water (Figure 5.11d).

At the surface of a cell, phospholipids are arranged in a similar bilayer. The hydrophilic heads of the molecules are on the outside of the bilayer, in contact with the aqueous solutions inside and outside of the cell. The hydrophobic tails point toward the interior of the bilayer, away from the water. The phospholipid bilayer forms a boundary between the cell and its external environment and establishes separate compartments within eukaryotic cells; in fact, the existence of cells depends on the properties of phospholipids.

Steroids

Steroids are lipids characterized by a carbon skeleton consisting of four fused rings. Different steroids are distinguished by the particular chemical groups attached to this ensemble of rings. **Cholesterol**, a type of steroid, is a crucial molecule in animals **(Figure 5.12)**. It is a common component of animal cell membranes and is also the precursor from which other steroids, such as the vertebrate sex hormones, are synthesized. In vertebrates, cholesterol is synthesized in the liver and is also obtained from the diet. A high level of cholesterol in the blood may contribute to atherosclerosis, although some researchers are questioning the roles of cholesterol and saturated fats in the development of this condition.

Figure 5.12

Cholesterol, a steroid.

A molecule.

Cholesterol is the molecule from which other steroids, including the sex hormones, are synthesized. Steroids vary in the chemical groups attached to their four interconnected rings (shown in gold).

Cholesterol contains the four fused rings of a steroid. It has an alcohol on the A ring, a branched 8-carbon tail on the D ring, and a methyl group at both the AB and CD ring junctions. There is also one double bond in the B ring. All other steroids are made from cholesterol.

MAKE CONNECTIONS *Compare cholesterol with the sex hormones shown in the figure at the beginning of*Concept 4.3*. Circle the chemical groups that cholesterol has in common with estradiol; put a square around the chemical groups that cholesterol has in common with testosterone.*

*For suggested answer, see**Appendix A.*

Interview

Interview with Lovell Jones: Investigating the effects of sex hormones on cancer

A photo of Lovell Jones.

Concept Check 5.3

- Compare the structure of a fat (triglyceride) with that of a phospholipid.
- Why are human sex hormones considered lipids?
- **WHAT IF?** Suppose a membrane surrounded an oil droplet, as it does in the cells of plant seeds and in some animal cells. Describe and explain the form it might take.

*For suggested answers, see*Appendix A.*

Concept 5.4: Proteins include a diversity of structures, resulting in a wide range of functions

Nearly every dynamic function of a living being depends on proteins. In fact, the importance of proteins is underscored by their name, which comes from the Greek word *proteios*, meaning “first,” or “primary.” Proteins account for more than 50% of the dry mass of most cells, and they are instrumental in almost everything organisms do. Some proteins speed up chemical reactions, while others play a role in defense,

storage, transport, cellular communication, movement, or structural support. **Figure 5.13** shows examples of proteins with these functions, which you'll learn more about in later chapters.

Figure 5.13

An overview of protein functions.

An overview of protein functions.

Storage proteins. Function: Storage of amino acids. Examples: Casein, the protein of milk, is the major source of amino acids for baby mammals. Plants have storage proteins in their seeds. Ovalbumin is the protein of egg white, used as an amino acid source for the developing embryo.

Hormonal proteins. Vessel with high blood sugar levels causes insulin to be secreted, resulting in vessel with normal blood sugar levels. Function: Coordination of an organism's activities. Example: Insulin, a hormone secreted by the pancreas, causes other tissues to take up glucose, thus regulating blood sugar concentration.

Contractile and motor proteins. Function: Movement. Examples: Motor proteins are responsible for the undulations of cilia and flagella. Actin and myosin proteins are responsible for the contraction of muscles.

Defensive proteins. Function: Protection against disease. Example: Antibodies inactivate and help destroy viruses and bacteria.

Transport proteins. A transport protein imbedded in the cell membrane allows for the passage of a substance into the cell. Function: Transport of substances. Examples: Hemoglobin, the iron-containing protein of vertebrate blood, transports oxygen from the lungs to other parts of the body. Other proteins transport molecules across membranes, as shown here.

Receptor proteins. A nerve cell releases signaling molecules that bind to receptors on the surface of another nerve cell. Function: Response of cell to chemical stimuli. Example: Receptors built into the membrane of a nerve cell detect signaling molecules released by other nerve cells.

Structural proteins. Function: Support. Examples: Keratin is the protein of hair, horns, feathers, and other skin appendages. Insects and spiders use silk fibers to make their cocoons and webs, respectively. Collagen and elastin proteins provide a fibrous framework in animal connective tissues.

Animation

Protein Functions

image

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image

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Life would not be possible without enzymes, most of which are proteins. Enzymatic proteins regulate metabolism by acting as **catalysts**, chemical agents that selectively speed up chemical reactions without being consumed in the reaction. Because an enzyme can perform its function over and over again, these molecules can be thought of as workhorses that keep cells running by carrying out the processes of life.

A human has tens of thousands of different proteins, each with a specific structure and function; proteins, in fact, are the most structurally sophisticated molecules known. Consistent with their diverse functions, they vary extensively in structure, each type of protein having a unique three-dimensional shape.

Proteins are all constructed from the same set of 20 amino acids, linked in unbranched polymers. The bond between amino acids is called a *peptide bond*, so a polymer of amino acids is called a **polypeptide**. A **protein** is a biologically functional molecule made up of one or more polypeptides, each folded and coiled into a specific three-dimensional structure.

Amino Acids (Monomers)

All amino acids share a common structure. An **amino acid** is an organic molecule with both an amino group and a carboxyl group (see Figure 4.9); the small figure shows the general formula for an amino acid. At the center of the amino acid is an asymmetric carbon atom called the *alpha* α *carbon*. Its four different partners are an amino group, a carboxyl group, a hydrogen atom, and a variable group symbolized by R. The R group, also called the side chain, differs with each amino acid. The R group may be as simple as a hydrogen atom, or it may be a carbon skeleton with various functional groups attached. The physical and chemical properties of the side chain determine the unique characteristics of a particular amino acid, thus affecting its functional role in a polypeptide.

Structure of an amino acid.

A central carbon (alpha carbon) is linked to an amino group (N H 2), a carboxyl group (C double bonded to O and single bond to O H), an H, and an R group (side chain).

Figure 5.14 shows the 20 amino acids that cells use to build their thousands of proteins. Here the amino groups and carboxyl groups are all depicted in ionized form, the way they usually exist at the pH found in a cell. The amino acids are grouped according to the properties of their side chains. One group consists of amino acids with nonpolar side chains, which are hydrophobic. Another group consists of amino acids with polar side chains, which are hydrophilic. Acidic amino acids have side chains that are generally negative in

charge due to the presence of a carboxyl group, which is usually dissociated (ionized) at cellular pH. Basic amino acids have amino groups in their side chains that are generally positive in charge. (The terms *acidic* and *basic* in this context refer only to groups in the side chains because *all* amino acids—as monomers—have carboxyl groups and amino groups.) Because they are charged, acidic and basic side chains are also hydrophilic.

Figure 5.14

The 20 amino acids of proteins.

Detailed structures and abbreviations for the 20 natural amino acids.

The 20 natural amino acids are grouped here according to the properties of their side chains (R groups) and shown in their prevailing form at pH 7.2, the pH within a cell. The three-letter and one letter abbreviations for each amino acid are given in parenthesis. Each of the amino acids has the same general structure, namely, H3N+–CHR–COO-. The R group is the only part that differs. Throughout this description, groups of atoms that branch off the main chain of the R group are set off with parenthesis in the formula. All of the amino acids used in proteins are L enantiomers (see Figure 4.7c).

a. Nonpolar (Hydrophobic) Side Chains.

- glycine (Gly or G): R = single bond H.
- alanine (Ala or A): R = single bond C H 3 (methyl).
- valine (Val or V): R = single bond C H, C H 3, 2, (isopropyl).
- leucine (Leu or L): R = single bond C H 2. single bond, C H, C H 3, 2 (isobutyl).
- isoleucine (Ile or I): R = single bond C H, C H 3, single bond, C H 2, single bond, C H 3 (sec-butyl).
- methionine (Met or M): R = single bond C H 2, single bond, C H 2, single bond, S, single bond, C H 3.
- phenylalanine (Phe or F): R = single bond, C H 2, single bond, C 6 H 5 (benzyl). This side chain contains an aromatic ring.
- tryptophan (Trp or W): R = single bond C H 2, single bond, indole. The R group for this amino acid is relatively complex: Its CH2 group is bonded to the number 3 atom of an indole molecule, which consists of fused five- and six-membered rings. To draw indole, start with a five-membered ring containing an NH group at position 1. Draw double bonds between carbons 2 and 3 and between carbons 4 and 5. Carbon 3 is bonded to the CH2 leading to the backbone of the amino acid. Complete the indole molecule, using carbons 4 and 5 as the basis for forming the six-membered ring. Make the six-membered ring aromatic by including two more double bonds.
- proline (Pro or P): R = single bond, C H 2, single bond, CH 2, single bond, CH 2, single bond. Proline is unique, as it is the only amino acid where the R group wraps around to forms a ring with the backbone (specifically, the amino group) of the amino acid. This makes the backbone have the

slightly different form R N H 2 plus C H R, single bond, C O 2 plus, where both R's refers to the same three carbon bridge).

b. Polar (Hydrophilic) Side Chains:

- serine (Ser or S): R = single bond C H 2, single bond, O H.
- threonine (Thr or T): R = single bond, C H, O H, single bond, C H 3.
- cysteine (Cys or C): R = single bond C H2, single bond, S H.
- tyrosine (Tyr or Y): R = single bond C H2, single bond, C 6 H 4, single bond, O H (Tyrosine contains an aromatic ring. The single bond O H is para to the amino acid backbone.)
- asparagine (Asn or N): R = single bond C H2, single bond, C O, single bond, N H 2 (contains a carbonyl, amide).
- glutamine (Gln or Q): R = single bond, C H 2, single bond, C H 2, single bond, C O, single bond, N H 2 (contains a carbonyl, amide.

c. Electrically charged side chains (hydrophilic), Acidic, negatively charged:

- aspartic acid (Asp or D): R = single bond, C H 2, single bond, C O O- (contains a carboxylate ion in neutral solutions).
- glutamic acid (Glu or E): R = single bond, C H 2, single bond, C H 2, single bond, C O O- (contains a carboxylate ion in neutral solutions).

d. Electrically charged side chains (hydrophilic), Basic, Positively Charged:

- lysine (Lys or K): R = –single bond, C H 2, single bond, C H 2, single bond, C H 2, single bond, C H 2, single bond, N H 3 plus.
- arginine (Arg or R): R = single bond, C H 2, single bond, C C H 2, single bond, C H 2, single bond, N H, single bond, C, N H 2, plus, single bond, N H 2. The NH 2 plus is double-bonded to the carbon.
- histidine (His or H): R = single bond, C H 2, single bond, imidazole. Histidine has a relatively complex structure. To draw it, begin with the structure of imidazole, a five-membered ring containing a nitrogen atom at both positions 1 and 3. Add a hydrogen atom to both nitrogen atoms, and a plus charge to nitrogen 3. Draw a double bond between carbon 2 and nitrogen 3, and another double bond between carbons 4 and 5. Carbon 5 bonds to the CH2 leading back to the amino acid backbone.

At the top right are the non-ionized and ionized forms of a generic amino acid. The specific amino acids are shown below in their ionized form, prevalent at the pH within a cell (pH 7.2). The amino acids are grouped by properties of their side chains. The three-letter and one-letter abbreviations for the amino acids are in parentheses. All of the amino acids used in proteins are L enantiomers (see Figure 4.7c).

Polypeptides (Amino Acid Polymers)

Now that we have examined amino acids, let's see how they are linked to form polymers **(Figure 5.15)**. When two amino acids are positioned so that the carboxyl group of one is adjacent to the amino group of the other, they can become joined by a dehydration reaction, with the removal of a water molecule. The resulting covalent bond is called a **peptide bond**. Repeated over and over, this process yields a polypeptide, a polymer of many amino acids linked by peptide bonds. You'll learn more about how cells synthesize polypeptides in Concept 17.4.

Figure 5.15

Making a polypeptide chain.

The structure and synthesis of a polypeptide chain.

The polypeptide backbone consists of a series of amino acids held together with peptide bonds. The left end of the backbone as drawn is called the amino end (N-terminus). Beginning here, the backbone is series of repeating N-C-C amino acid units. Each amino acid has a side chain on its middle (alpha) carbon. The carboxyl carbon of each amino acid forms a peptide bond with the nitrogen of the next amino acid. The right-hand end of the polypeptide is called the carboxyl end (C-terminus). New amino acids are added one at a time to the carboxyl end of the peptide. With each new amino acid, a dehydration reaction makes a water molecule from the O H of the peptide's carboxyl end and an H atom from the amino group of the new amino acid. A new peptide bond is also formed between the carboxyl and amino groups.

Peptide bonds are formed by dehydration reactions, which link the carboxyl group of one amino acid to the amino group of the next. The peptide bonds are formed one at a time, starting with the amino acid at the amino end (N-terminus). The polypeptide has a repetitive backbone (purple) to which the amino acid side chains (yellow and green) are attached.

DRAW IT *Label the three amino acids in the upper part of the figure using three-letter and one-letter codes. Circle and label the carboxyl and amino groups that will form the new peptide bond.*

*For suggested answer, see**Appendix A.*

The repeating sequence of atoms highlighted in purple in Figure 5.15 is called the *polypeptide backbone*. Extending from this backbone are the different side chains (R groups) of the amino acids. Polypeptides range in length from a few amino acids to 1,000 or more. Each specific polypeptide has a unique linear sequence of amino acids. Note that one end of the polypeptide chain has a free amino group (the N-terminus of the polypeptide), while the opposite end has a free carboxyl group (the C-terminus). The chemical nature of the molecule as a whole is determined by the kind and sequence of the side chains, which determine how a polypeptide folds and thus its final shape and chemical characteristics. The immense variety of polypeptides

in nature illustrates an important concept introduced earlier—that cells can make many different polymers by linking a limited set of monomers into diverse sequences.

Protein Structure and Function

The specific activities of proteins result from their intricate three-dimensional architecture, the simplest level of which is the sequence of their amino acids. What can the amino acid sequence of a polypeptide tell us about the three-dimensional structure (commonly referred to simply as the "structure") of the protein and its function? The term *polypeptide* is not synonymous with the term *protein*. Even for a protein consisting of a single polypeptide, the relationship is somewhat analogous to that between a long strand of yarn and a sweater of particular size and shape that can be knitted from the yarn. A functional protein is not *just* a polypeptide chain, but one or more polypeptides precisely twisted, folded, and coiled into a molecule of unique shape, which can be shown in several different types of models **(Figure 5.16)**. And it is the amino acid sequence of each polypeptide that determines what three-dimensional structure the protein will have under normal cellular conditions.

Advanced placement edition

Apply concepts and connect across Big Ideas using visual representations.

What determines the properties of a protein? What structural properties are critical for membrane-bound transport proteins? How is a change in a protein and gene expression related to a change in DNA sequence?

Figure 5.16

Visualizing Proteins

Visualizing proteins.

Proteins can be represented in different ways, depending on the goal of the illustration.

Structural models: Using data from structural studies of proteins, computers can generate various types of models. Each model emphasizes a different aspect of the protein's structure, but no model can show what a protein actually looks like. These three models depict lysozyme, a protein in tears and saliva that helps prevent infection by binding to target molecules on bacteria. From left to right, these models are space-filling, ribbon, and wireframe.

- Space-filling model: Shows all the atoms of the protein (except hydrogen), emphasizing the overall globular shape. The atoms are color-coded: gray = carbon, red = oxygen, blue = nitrogen, and yellow = sulfur. Groove is an indent in the model.

- Ribbon model: Shows only the backbone of the polypeptide, emphasizing how it folds and coils to form a 3-D shape, in this case stabilized by disulfide bridges (yellow lines). Groove is shown as an indent in the folded ribbon structure.
- Wireframe model (blue): Shows the backbone of the polypeptide chain with side chains (R groups) extending from it (see Figure 5.15). A ribbon model (purple) is superimposed on the wireframe model. Target molecule (on the bacterial cell surface) bound to lysozyme, shown joining with "groove" in model.

Simplified diagrams: It isn't always necessary to use a detailed computer J17model; simplified diagrams are useful when the focus of the figure is on the function of the protein, not the structure. In this diagram of the protein rhodopsin, a simple transparent shape is drawn around the contours of a ribbon model, showing the overall shape of the molecule as well as some internal details. When structural details are not needed, a solid shape can be used to represent a protein. Image shows protein as a purple oval shaded to look three-dimensional. A simple shape is used here to represent a generic enzyme because the diagram focuses on enzyme action in general. Image shows two enzymes, both shown as purple ovals. The first enzyme is shown with two green hexagonal food molecules which are linked to each other. An arrow flows to the second enzyme which has broken the bonds between the food molecules, and the molecules are shown moving away from the enzyme. Sometimes a protein is represented simply as a dot, as shown here for insulin. Image shows insulin-producing cell in pancreas. Cell is an orange oval with four small green dots on its left side representing insulin.

*For suggested answers, see**Appendix A*.

Instructors: The tutorial "Molecular Model: Lysozyme," in which students rotate 3-D models of lysozyme, can be assigned in **Mastering Biology**.

Instructors: Additional questions related to this Visualizing Figure can be assigned in **Mastering Biology**.

When a cell synthesizes a polypeptide, the chain may fold spontaneously, assuming the functional structure for that protein. This folding is driven and reinforced by the formation of various bonds between parts of the chain, which in turn depends on the sequence of amino acids. Many proteins are roughly spherical (*globular proteins*), while others are shaped like long fibers (*fibrous proteins*). Even within these broad categories, countless variations exist.

A protein's specific structure determines how it works. In almost every case, the function of a protein depends on its ability to recognize and bind to some other molecule. In an especially striking example of the marriage of form and function, **Figure 5.17** shows the exact match of shape between an antibody (a protein in the body) and the particular foreign substance on a flu virus that the antibody binds to and marks for destruction. Also, you may recall another example of molecules with matching shapes from Concept 2.3: endorphin molecules (produced by the body) and morphine molecules (a manufactured drug), both of which

fit into receptor proteins on the surface of brain cells in humans, producing euphoria and relieving pain. Morphine, heroin, and other opiate drugs are able to mimic endorphins because they all have a shape similar to that of endorphins and can thus fit into and bind to endorphin receptors in the brain. This fit is very specific, something like a handshake (see Figure 2.16). The endorphin receptor, like other receptor molecules, is a protein. The function of a protein—for instance, the ability of a receptor protein to bind to a particular pain-relieving signaling molecule—is an emergent property resulting from exquisite molecular order.

Figure 5.17

Complementarity of shape between two protein surfaces.

Different parts of an antibody protein and a protein from the flu virus are colored. The protein and virus have complementary shapes in parts.

A technique called X-ray crystallography was used to generate a computer model of an antibody protein (blue and orange, left) bound to a flu virus protein (yellow and green, right). This is a wireframe model modified by adding an "electron density map" in the region where the two proteins meet. Computer software was then used to back the images away from each other slightly.

VISUAL SKILLS *What do these computer models allow you to see about the two proteins?*

*For suggested answer, see**Appendix A.*

Four Levels of Protein Structure

In spite of their great diversity, proteins share three superimposed levels of structure, known as **primary structure**, **secondary structure**, and **tertiary structure**. A fourth level, quaternary structure, arises when a protein consists of two or more polypeptide chains. **Figure 5.18** describes these four levels of protein structure. Be sure to study this figure thoroughly before going on to the next section.

Animation

Protein Structure

image

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Figure 5.18

Exploring Levels of Protein Structure

Exploring the levels of protein structure.

The primary structure of a protein is its sequence of amino acids. As an example, let's consider transthyretin, a globular blood protein that transports vitamin A and one of the thyroid hormones throughout the body. Transthyretin is made up of four identical polypeptide chains, each composed of 127 amino acids. Shown here is one of these chains unraveled for a closer look at its primary structure. Each of the 127 positions along the chain is occupied by one of the 20 amino acids, indicated here by its three-letter abbreviation. The primary structure is like the order of letters in a very long word. If left to chance, there would be 20127 different ways of making a polypeptide chain 127 amino acids long. However, the precise primary structure of a protein is determined not by the random linking of amino acids, but by inherited genetic information. The primary structure in turn dictates secondary and tertiary structure, due to the chemical nature of the backbone and the side chains (R groups) of the amino acids along the polypeptide. Each amino acid is made up of a central carbon attached to an amino group, a carboxyl group, an H and a side group (one of 20 amino acids).

Secondary structure: Regions stabilized by hydrogen bonds between atoms of the polypeptide backbone. Includes alpha helices and beta pleated sheets with structures supported by hydrogen bonds. Most proteins have segments of their polypeptide chains repeatedly coiled or folded in patterns that contribute to the protein's overall shape. These coils and folds, collectively referred to as secondary structure, are the result of hydrogen bonds between the repeating constituents of the polypeptide backbone (not the amino acid side chains). Within the backbone, the oxygen atoms have a partial negative charge, and the hydrogen atoms attached to the nitrogens have a partial positive charge (see Figure 2.14); therefore, hydrogen bonds can form between these atoms. Individually, these hydrogen bonds are weak, but because they are repeated many times over a relatively long region of the polypeptide chain, they can support a particular shape for that part of the protein. One such secondary structure is the alpha helix, a delicate coil held together by hydrogen bonding between every fourth amino acid, as shown above. Although each transthyretin polypeptide has only one alpha helix region (see the Tertiary Structure section), other globular proteins have multiple stretches of alpha helix separated by nonhelical regions (see hemoglobin in the Quaternary Structure section). Some fibrous proteins, such as alpha-keratin, the structural protein of hair, have the alpha helix formation over most of their length. The other main type of secondary structure is the beta pleated sheet. As shown above, in this structure two or more segments of the polypeptide chain lying side by side (called beta strands) are connected by hydrogen bonds between parts of the two parallel segments of polypeptide backbone. Beta pleated sheets make up the core of many globular proteins, as is the case for transthyretin (see Tertiary Structure), and dominate some fibrous proteins, including the silk protein of a spider's web. The teamwork of so many hydrogen bonds makes each spider silk fiber stronger than a steel

strand of the same weight. Spiders secrete silk fibers made of a structural protein containing beta pleated sheets, which allow the spider web to stretch and recoil.

Tertiary structure: three-dimensional shape stabilized by interactions between side chains. Diagram shows ribbon model of transthyretin polypeptide, which includes regions of both alpha helices and beta pleated sheets. Superimposed on the patterns of secondary structure is a protein's tertiary structure, shown here in a ribbon model of the transthyretin polypeptide. While secondary structure involves interactions between backbone constituents, tertiary structure is the overall shape of a polypeptide resulting from interactions between the side chains (R groups) of the various amino acids. One type of interaction that contributes to tertiary structure is called, somewhat misleadingly, a hydrophobic interaction. As a polypeptide folds into its functional shape, amino acids with hydrophobic (nonpolar) side chains usually end up in clusters at the core of the protein, out of contact with water. Thus, a "hydrophobic interaction" is actually caused by the exclusion of nonpolar substances by water molecules. Once nonpolar amino acid side chains are close together, van der Waals interactions help hold them together. Meanwhile, hydrogen bonds between polar side chains and ionic bonds between positively and negatively charged side chains also help stabilize tertiary structure. These are all weak interactions in the aqueous cellular environment, but their cumulative effect helps give the protein a unique shape. Covalent bonds called disulfide bridges may further reinforce the shape of a protein. Disulfide bridges form where two cysteine monomers, which have sulfhydryl groups (S H) on their side chains (see Figure 4.9), are brought close together by the folding of the protein. The sulfur of one cysteine bonds to the sulfur of the second, and the disulfide bridge (single bond, S, single bond, S, single bond) rivets parts of the protein together (see yellow lines in Figure 5.16 ribbon model). All of these different kinds of interactions can contribute to the tertiary structure of a protein.

Quaternary Structure: Association of two or more polypeptides (some proteins only). Transthyretin protein is made up of four identical polypeptides, each with secondary, tertiary and primary structure. Some proteins consist of two or more polypeptide chains aggregated into one functional macromolecule. Quaternary structure is the overall protein structure that results from the aggregation of these polypeptide subunits. For example, shown above is the complete globular transthyretin protein, made up of its four polypeptides. Another example is collagen, which is a fibrous protein that has three identical helical polypeptides intertwined into a larger triple helix, giving the long fibers great strength. This suits collagen fibers to their function as the girders of connective tissue in skin, bone, tendons, ligaments, and other body parts. (Collagen accounts for 40% of the protein in a human body.)

Animation

Proteins

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MP3 Tutor

Protein structure and Function

Hey there, and welcome to another MP3 Tutor session. This is Eric Simon and I'll guide you through today's topic: proteins. In your text and in other MP3 sessions, you've been exposed to one of the major themes of biology: the relationship between structure and function. In biology, we say that form follows function. That is, the structure of an object reflects the job done. For example, the form of a chair must reflect its function. Chairs can come in many different shapes and sizes, but they all must have a flat part to sit on and a stable part to meet the floor. Throughout multiple levels of biological organization, exploring the theme of structure and function can help you to develop a deeper understanding of how life works. In this MP3 tutor session, we will explore how proteins are put together and relate that structure to a few specific examples of protein function. Sound good? Let's get started. When I hear the word "protein" the first thing that usually pops in my mind is food: meat, dairy products, eggs, that sort of thing. Of course, I think about food a lot. However, even when my appetite is saying "salad," I always make sure that my salad has ample protein in the form of nuts, seeds, and beans. Proteins in food provide the raw material to build the bodily proteins essential for your good health. Proteins have the most diverse and complex architecture of any of the biological macromolecules. You have tens of thousands of different proteins in your body that have diverse structures and perform diverse jobs. Proteins are the worker bees of your body. They take part in nearly every activity of an organism. Proteins play a role in structural support (as in cartilage), transport (such as hemoglobin), hormones (like insulin), immunity (your antibodies), movement (as with muscle proteins), and as enzymes that keep thousands of different chemical reactions running inside all cells. In each of these examples, the shape or structure of a protein is the key to understanding how it works. How can the shape of a protein be so important? Let's try an analogy. Pliers are one of the most commonly used tools for extending the power and grip of your hand. Most people have used pliers to accomplish a specific task, such as cracking a nut, removing a bolt, bending wire, cutting wire, and so on. In each case, the moveable jaws of the pliers are shaped for specific tasks. Using them for a different duty sometimes leads to frustration. For example, you can't remove a bolt with a pair of needle-nosed pliers. Shape is important to a tool's function, and the same is true for proteins. Each of the thousands of different kinds of proteins in your body has its own unique shape. Describing so many different proteins, with so many different shapes, sounds like a daunting task. Luckily, it's not as difficult as it sounds. Let's explore how proteins are put together. When describing the structure of proteins, biologists think about four distinct levels of protein structure. These four levels are known as primary, secondary, tertiary, and quaternary. I always have trouble pronouncing these words so sometimes I simply refer to these levels as the first, second, third or fourth level of protein structure. The first or primary level of protein structure is simply the linear sequence of amino acids that are strung together to make the

protein chain. Every protein has a unique amino acid sequence. Just as you can build thousands of words using just 26 letters of the English alphabet, a huge variety of proteins can be constructed from 20 different amino acids. Just like English letters have different properties-such as consonants vs. vowels-so do amino acids. The properties of an individual amino acid lay in its side chain that branches off of the carbon backbone. The side chain can also be called the R group or radical group. Let me use another analogy. Different railroad cars (such as tanker cars, flatbed cars, or refrigerator cars) can be built on a standard railroad car chassis so that cars can be easily linked together to form a long train. Likewise, a common carbon backbone is found in all amino acids. This common backbone permits amino acids to be linked together in long chains. Unlike a train, however, the chain of amino acids folds and twists to form a complex three dimensional shape. The shape eventually taken by a chain of amino acids is directly tied to the side groups that protrude from the amino acid chain. If even one amino acid is out of order it can make the entire protein non-functioning. However, let me point out that some amino acids can be substituted for others without altering the function of a protein. Let's go back to the word analogy for a second. If you change just one letter in some words you can completely change the meaning. For example, changing one letter in the word "ship" can produce a complete different word like "shop"-very different meanings, right? Other words, however, are not as susceptible to single changes. If you change a letter around, they may be misspelled, but their meaning may not be changed. For example, if you switch around the "e" and the "i" in the word "protein" you'd have no trouble understanding the meaning of the word. If the amino acid sequence of a protein is changed, most changes result in defective proteins but some do not. To recap, the primary structure of a protein is simply the list of amino acids in their proper order. To understand the secondary level of protein structure, imagine that you are an alien seeing a roomful of human beings for the first time. The first thing you might notice is that every person has a unique shape. But you would quickly start to notice similarities in the shapes. Every person, for example, has a rectangular body with a spherical blob at the top and four long limbs sticking out. If you look at many proteins, you will similarly notice certain common structural elements. The two most common are called an alpha helix and a beta sheet. An alpha helix is a region of a protein that is twisted into a tight spiral. The coils of the helix are held together by hydrogen bonds between nearby amino acids. Other regions of a protein fold up like a paper fan. This type of structure is referred to as a beta sheet. The alpha helices and the beta sheets together comprise the secondary structure of a protein. A given protein might have three alpha helices and four beta sheets, while another has eight and two, and so on. Most proteins have at least some of each. Once all the helices and sheets are coiled, folded, and locked into place, the polypeptide takes on a distinct three-dimensional shape. This is the third or tertiary level of protein structure: the overall conformation or three-dimensional shape of a protein. To think about protein shapes, imagine that I have a big barrel of clay and I ask everyone in the class to grab a lump and put it on their desk. At first glance, all the lumps look similar; they are all blobby lumps of clay. But closer observation reveals that every lump of clay is unique. Some are taller or fatter, some have bumps and grooves that others lack. Similarly, while most proteins look basically like round-ish blobs, each one has small differences in their structure. And these differences in shape are vital to the job a protein performs. That is, form follows function. Change a protein's shape in the slightest (by, say, cutting off a lump) and the

protein may not be able to perform its job within the cell. The fourth level of protein structure describes interactions among different polypeptides. Some complex proteins are composed of more than one polypeptide chain that bond together to form a sort of super protein. For example, the blood protein hemoglobin is composed of four different polypeptide chains that interlock. Another example is the molecular machine known as DNA polymerase, the enzyme that builds new DNA molecules during DNA replication. DNA polymerase is actually a large complex made up of a number of different polypeptide chains joined together. Not all proteins have quaternary structure, but the big ones usually do. As a summary, let's review the four levels of protein structure: The first level is the sequence of amino acids. The second level is the formation of alpha helices and beta sheets. The third level is the folding of a polypeptide into its final three dimensional shape. The fourth level is the joining of more than one chain to form a larger complex. I hope this gives you a leg up on this topic. If you need more help, please refer to your book, CD-ROM, or course website. Good luck with your studies.

Sickle-Cell Disease: A Change in Primary Structure

Even a slight change in primary structure can affect a protein's shape and ability to function. For instance, **sickle-cell disease**, an inherited blood disorder, is caused by the substitution of one amino acid (valine) for the normal one (glutamic acid) at the position of the sixth amino acid in the primary structure of hemoglobin, the protein that carries oxygen in red blood cells. Normal red blood cells are disk-shaped, but in sickle-cell disease, the abnormal hemoglobin molecules tend to aggregate into chains, deforming some of the cells into a sickle shape **(Figure 5.19)**. A person with the disease has periodic "sickle-cell crises" when the angular cells clog tiny blood vessels, impeding blood flow. The toll taken on such patients is a dramatic example of how a simple change in protein structure can have devastating effects on protein function.

Figure 5.19

A single amino acid substitution in a protein causes sickle-cell disease.

Normal and sickle-cell hemoglobin is compared.

The table is as follows.

Hemoglobin Type	Primary Structure	Secondary and Tertiary Structures	Quaternary Structure	Function	Red Blood Cell Shape
Normal hemoglobin	Sequence of amino acids: Val,	Normal beta subunit A	Normal alpha subunits and 2 normal beta	Normal hemoglobin proteins do	Normal red blood cells are full of

	His, Leu, Thr, Pro, Glu, Glu.	spherical, blue colored subunit is depicted.	Two spherical, blue-colored beta subunits are superimposed on purple, smaller alpha subunits.	not associate with one another; each carries oxygen. Small images of hemoglobin with blue and purple subunits are depicted.	individual hemoglobin molecules. An image of a normal red blood cell shows a disc shape (scale: 5 micrometers).
Sickle-cell hemoglobin	Sequence of amino acids: Val, His, Leu, Thr, Pro, Val, Glu. The second "Val" was a "Glu" in normal hemoglobin.	Sickle-cell beta subunit is malformed —has a slight "sickle" shape as compared to the normal beta subunit. A spherical, red-colored beta subunit with a small indentation on the left side is shown.	In sickle-cell hemoglobin, both beta subunits have a slight sickle shape. Both alpha subunits are normal. A hemoglobin molecule with 2 abnormal beta subunits in red, superimposed on two normal alpha subunits is shown.	Hydrophobic interactions between sickle-cell hemoglobin proteins lead to their aggregation into a fiber; the capacity to carry oxygen is greatly reduced. A fiber of hemoglobin is shown, which upon magnification shows sickle-cell, abnormal hemoglobin molecules alternately	Fibers of abnormal hemoglobin deform red blood cells into a sickle shape. Image of a sickle-shaped red blood cell (scale: 5 micrometers).

				stacked to form a fiber.	

MAKE CONNECTIONS *Considering the chemical characteristics of the amino acids valine and glutamic acid (see*Figure 5.14*), propose a possible explanation for the dramatic effect on protein function that occurs when valine is substituted for glutamic acid.**

*For suggested answer, see**Appendix A*.

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Sickle-Cell Disease

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Interview

Interview with Linus Pauling: Winner of the Nobel Prize in Chemistry and the Nobel Peace Prize

A photo of Linus Pauling.

What Determines Protein Structure?

You've learned that a unique shape endows each protein with a specific function. But what are the key factors determining protein structure? You already know most of the answer: A polypeptide chain of a given amino acid sequence can be arranged into a three-dimensional shape determined by the interactions responsible for secondary and tertiary structure. This folding normally occurs as the protein is being synthesized in the crowded environment within a cell, aided by other proteins. However, protein structure also depends on the physical and chemical conditions of the protein's environment. If the pH, salt concentration, temperature, or other aspects of its environment are altered, the weak chemical bonds and interactions within a protein may be destroyed, causing the protein to unravel and lose its native shape, a change called **denaturation(Figure 5.20)**. Because it is misshapen, the denatured protein is biologically inactive.

Figure 5.20

Denaturation and renaturation of a protein.

A normal protein goes through denaturation to become a denatured protein. A denatured protein goes through renaturation to become a normal protein.

High temperatures or various chemical treatments will denature a protein, causing it to lose its shape and hence its ability to function. If the denatured protein remains dissolved, it may renature when the chemical and physical aspects of its environment are restored to normal.

See table next tab

Most proteins become denatured if they are transferred from an aqueous environment to a nonpolar solvent, such as ether or chloroform; the polypeptide chain refolds so that its hydrophobic regions face outward toward the solvent. Other denaturation agents include chemicals that disrupt the hydrogen bonds, ionic bonds, and disulfide bridges that maintain a protein's shape. Denaturation can also result from excessive heat, which agitates the polypeptide chain enough to overpower the weak interactions that stabilize the structure. The white of an egg becomes opaque during cooking because the denatured proteins are insoluble and solidify. This also explains why excessively high fevers can be fatal: Proteins in the blood tend to denature at very high body temperatures.

When a protein in a test-tube solution has been denatured by heat or chemicals, it can sometimes return to its functional shape when the denaturing agent is removed. (Sometimes this is not possible: For example, a fried egg will not become liquefied when placed back into the refrigerator!) We can conclude that the information for building specific shape is intrinsic to the protein's primary structure; this is often the case for small proteins. The sequence of amino acids determines the protein's shape—where an **alpha helix can form, where betapleated sheets** can exist, where disulfide bridges are located, where ionic bonds can form, and so on. But how does protein folding occur in the cell?

Protein Folding in the Cell

Biochemists now know the amino acid sequence for over a hundred million proteins, with several million added each month, and the three-dimensional shape for tens of thousands. Researchers have tried to correlate the primary structure of many proteins with their three-dimensional structure to discover the rules of protein folding. Unfortunately, however, the protein-folding process is not that simple. Most proteins probably go through several intermediate structures on their way to a stable shape, and looking at the mature structure does not reveal the stages of folding required to achieve that form. However, biochemists have developed methods for tracking a protein through such stages and learning more about this important process.

Misfolding of polypeptides in cells is a serious problem that has come under increasing scrutiny by medical researchers. Many diseases—such as cystic fibrosis, Alzheimer's, Parkinson's, and mad cow disease—are associated with an accumulation of misfolded proteins. In fact, misfolded versions of the transthyretin protein featured in Figure 5.18 have been implicated in several diseases, including one form of senile dementia.

Even when scientists have a correctly folded protein in hand, determining its exact three-dimensional structure is not simple, for a single protein has thousands of atoms. The method most commonly used to determine the 3-D structure of a protein is **X-ray crystallography**, which depends on the diffraction of an X-ray beam by the atoms of a crystallized molecule. Using this technique, scientists can build a 3-D model that shows the exact position of every atom in a protein molecule **(Figure 5.21)**. Nuclear magnetic resonance (NMR) spectroscopy, cryo-electron microscopy (cryo-EM; see Concept 6.1) and bioinformatics (see Concept 5.6) are complementary approaches to understanding protein structure and function.

Figure 5.21

Research Method: X-Ray Crystallography

Application

Scientists use X-ray crystallography to determine the three-dimensional 3-D structure of macromolecules such as nucleic acids and proteins.

Technique

Researchers aim an X-ray beam through a crystallized protein or nucleic acid. The atoms of the crystal diffract (bend) the X-rays into an orderly array that a digital detector records as a pattern of spots called an X-ray diffraction pattern, an example of which is shown here.

X-ray crystallography.

Application: Scientists use X-ray crystallography to determine the three-dimensional structure of macromolecules such as nucleic acids and proteins.

Technique. Researchers aim an x-ray beam through crystalized protein or nucleic acid. The atoms of the crystal diffract (bend) the x-rays into an orderly array that a digital detector records as a pattern of spots called x-ray diffraction pattern, an example of which is given as the following diagram. An x-ray source sends a beam through a crystal. The beam scatters and hits a digital detector. A pattern emerges.

Results. Using data from x-ray diffraction patterns and the sequence of monomers determined by chemical methods, researchers can build a 3-D computer model of the macromolecule being studied, such as the four-subunit protein transthyretin.

Results

Using data from X-ray diffraction patterns and the sequence of monomers determined by chemical methods, researchers can build a 3-D computer model of the macromolecule being studied, such as the four-subunit protein transthyretin (see Figure 5.18) shown here.

A 3 D model of transthyretin, with 4 subunits, each colored differently.

The structure of some proteins is difficult to determine for a simple reason: A growing body of biochemical research has revealed that a significant number of proteins, or regions of proteins, do not have a distinct 3-D structure until they interact with a target protein or other molecule. Their flexibility and indefinite structure are important for their function, which may require binding with different targets at different times. These proteins, which may account for 20–30% of mammalian proteins, are called *intrinsically disordered proteins* and are the focus of current research.

Concept Check 5.4

- What parts of a polypeptide participate in the bonds that hold together secondary structure? Tertiary structure?
- Thus far in the chapter, the Greek letters α and β have been used to specify at least three different pairs of structures. Name and briefly describe them.
- Each amino acid has a carboxyl group and an amino group. Are these groups present in a polypeptide? Explain.
- **WHAT IF?** Where would you expect a polypeptide region rich in the amino acids valine, leucine, and isoleucine to be located in a folded polypeptide? Explain.

*For suggested answers, see*Appendix A.**

Concept 5.5: Nucleic acids store, transmit, and help express hereditary information

If the primary structure of polypeptides determines a protein's shape, what determines primary structure? The amino acid sequence of a polypeptide is programmed by a discrete unit of inheritance known as a

****gene. Genes consist of DNA, which belongs to the class of compounds called nucleic acids. **Nucleic acids** are polymers made of monomers called nucleotides.

The Roles of Nucleic Acids

The two types of nucleic acids, ********deoxyribonucleic acid (DNA)** and **ribonucleic acid (RNA)**, enable living organisms to reproduce their complex components from one generation to the next. Unique among molecules, DNA provides directions for its own replication. DNA also directs RNA synthesis and, through RNA, controls protein synthesis; this entire process is called **gene expression (Figure 5.22)**.

Figure 5.22

Gene expression: DNA→RNA→protein.

Gene expression.

In a eukaryotic cell, DNA in the nucleus programs protein production in the cytoplasm by dictating synthesis of messenger RNA (mRNA).

Three steps are given.

- 1. Messenger R N A (single stranded nucleic acid) is synthesized from the D N A (double stranded nucleic acid) in the nucleus.
- 2. m R N A leaves the nucleus via the nuclear pore and enters the cytoplasm.
- 3. m R N A enters a ribosome, where the information it carries is used to synthesize a protein.

The growing polypeptide chain consists of a chain of amino acids that were present separately in the cytoplasm before protein synthesis.

BioFlix® Animation

Gene Expression

image

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DNA is the genetic material that organisms inherit from their parents. Each chromosome contains one long DNA molecule, usually carrying several hundred or more genes. When a cell reproduces itself by dividing, its DNA molecules are copied and passed along from one generation of cells to the next. The information that programs all the cell's activities is encoded in the structure of the DNA. The DNA, however, is not directly involved in running the operations of the cell, any more than computer software by itself can read the bar code on a box of cereal. Just as a scanner is needed to read a bar code, proteins are required to implement genetic programs. The molecular hardware of the cell—the tools that carry out biological functions—consists mostly of proteins. For example, the oxygen carrier in red blood cells is the protein hemoglobin that you saw earlier (see Figure 5.18), not the DNA that specifies its structure.

How does RNA, the other type of nucleic acid, fit into gene expression, the flow of genetic information from DNA to proteins? A given gene along a DNA molecule can direct synthesis of a type of RNA called *messenger RNA* (*mRNA*). The mRNA molecule interacts with the cell's protein-synthesizing machinery to direct production of a polypeptide, which folds into all or part of a protein. We can summarize the flow of genetic information as DNA→RNA→protein (see Figure 5.22). The sites of protein synthesis are cellular structures called ribosomes. In a eukaryotic cell, ribosomes are in the cytoplasm—the region between the nucleus and the cell's outer boundary, the plasma membrane—but DNA resides in the nucleus. Messenger RNA conveys genetic instructions for building proteins from the nucleus to the cytoplasm. Prokaryotic cells lack nuclei but still use mRNA to convey a message from the DNA to ribosomes and other cellular equipment that translate the coded information into amino acid sequences. Later, you'll read about other functions of some recently discovered RNA molecules; the stretches of DNA that direct synthesis of these RNAs are also considered genes (see Concept 18.3).

The Components of Nucleic Acids

Nucleic acids are macromolecules that exist as polymers called **polynucleotides(Figure 5.23a)**. As indicated by the name, each polynucleotide consists of monomers called **nucleotides**. A nucleotide, in general, is composed of three parts: a five-carbon sugar (a pentose), a nitrogen-containing (nitrogenous) base, and one to three phosphate groups **(Figure 5.23b)**. The beginning monomer used to build a polynucleotide has three phosphate groups, but two are lost during the polymerization process. The portion of a nucleotide without any phosphate groups is called a *nucleoside*.

Figure 5.23

Components of nucleic acids.

Components of nucleic acids: Polynucleotides (nucleic acid), nucleotides, nucleosides, sugars, and nitrogenous bases (pyrimidines and purines).

Part a. Diagram of a polynucleotide, or nucleic acid, drawn from the 5 prime to 3 prime end. The sugar-phosphate backbone (alternates P-S-P-S) has nitrogenous bases attached to each sugar.

Part b. Structure of a nucleotide monomer in a polynucleotide. A phosphate group is attached to a five-carbon sugar (pentose), which is attached to a nitrogenous base. A nucleoside is the sugar plus nitrogenous base without the phosphate group.

Part c. Nucleoside components include the pyrimidines (the single-ring nitrogenous bases cytosine (C), thymine (T, in D N A) and uracil (U, in R N A)), the purines (double-ring nitrogenous bases adenine (A) and guanine (G)), and the sugars (deoxyribose in D N A and ribose in R N A). Both are five-carbon sugars, but deoxyribose contains one less oxygen atom than ribose.

(a) A polynucleotide has a sugarphosphate backbone with variable appendages, the nitrogenous bases. **(b)** In a polynucleotide, each nucleotide monomer includes a nitrogenous base, a sugar, and a phosphate group. Note that carbon numbers in the sugar include primes ′. **(c)** A nucleoside includes a nitrogenous base (purine or pyrimidine) and a fivecarbon sugar (deoxyribose or ribose).

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The Chemical Structure of DNA

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To understand the structure of a single nucleotide, let's first consider the nitrogenous bases **(Figure 5.23c)**. Each nitrogenous base has one or two rings that include nitrogen atoms. (They are called nitrogenous *bases* because the nitrogen atoms tend to take up H+ from solution, thus acting as bases.) There are two families of nitrogenous bases: pyrimidines and purines. A **pyrimidine** has one six-membered ring of carbon and nitrogen atoms. The members of the pyrimidine family are cytosine (C), thymine (T), and uracil (U). **Purines** are larger, with a six-membered ring fused to a five-membered ring. The purines are adenine (A) and guanine (G). The specific pyrimidines and purines differ in the chemical groups attached to the rings. Adenine, guanine, and cytosine are found in both DNA and RNA; thymine is found only in DNA and uracil only in RNA.

Now let's add the sugar to which the nitrogenous base is attached. In DNA the sugar is **deoxyribose**; in RNA it is **ribose** (see Figure 5.23c). The only difference between these two sugars is that deoxyribose lacks an oxygen atom on the second carbon in the ring, hence the name *deoxy*ribose.

So far, we have built a nucleoside (base plus sugar). To complete the construction of a nucleotide, we attach one to three phosphate groups to the 5′ carbon of the sugar (the carbon numbers in the sugar include′, the prime symbol; see Figure 5.23b). With one phosphate, this is a nucleoside monophosphate, more often called a nucleotide.

Nucleotide Polymers

The linkage of nucleotides into a polynucleotide involves a condensation reaction. (You will learn the details in Concept 16.2.) In the polynucleotide, adjacent nucleotides are joined by a phosphodiester linkage, which consists of a phosphate group that covalently links the sugars of two nucleotides. This bonding results in a repeating pattern of sugar-phosphate units called the *sugar-phosphate backbone* (see Figure 5.23a). (Note that the nitrogenous bases are not part of the backbone.) The two free ends of the polymer are distinctly different from each other. One end has a phosphate attached to a 5′ carbon, and the other end has a hydroxyl group on a 3′ carbon; we refer to these as the 5′*end* and the 3′*end*, respectively. We can say that a polynucleotide has a built-in directionality along its sugar-phosphate backbone, from 5′ to 3′, somewhat like a one-way street. The bases are attached all along the sugar-phosphate backbone.

The sequence of bases along a DNA (or mRNA) polymer is unique for each gene and provides very specific information to the cell. Because genes are hundreds to thousands of nucleotides long, the number of possible base sequences is effectively limitless. The information carried by the gene is encoded in its specific sequence of the four DNA bases. For example, the sequence 5′-AGGTAACTT-3′ means one thing, whereas the sequence 5′-CGCTTTAAC-3′ has a different meaning. (Entire genes, of course, are much longer.) The linear order of bases in a gene specifies the amino acid sequence—the primary structure—of a protein, which in turn specifies that protein's 3-D structure, thus enabling its function in the cell.

The Structures of DNA and RNA Molecules

A DNA molecule has two polynucleotides, or "strands," that wind around an imaginary axis, forming a **double helix(Figure 5.24a)**. The two sugar-phosphate backbones run in opposite 5′→3′ directions from each other; this arrangement is referred to as **antiparallel**, somewhat like a divided highway. The sugar-phosphate backbones are on the outside of the helix, and the nitrogenous bases are paired in the interior of the helix. The two strands are held together by hydrogen bonds between the paired bases (see Figure 5.24a). Most DNA molecules are very long, with thousands or even millions of base pairs. The one long DNA double helix in a eukaryotic chromosome includes many genes, each one a particular segment of the double-stranded molecule.

Figure 5.24

The structures of DNA and tRNA molecules.

The structure of D N A and t R N A molecules.

Each molecule has sugar-phosphate backbones and hydrogen bonds.

Part a. D N A has a double helix with a 3 prime strand and 5 prime strand. These ends reverse from one end of the molecule to the other. Base pairs are joined by hydrogen bonding.

Part b. Transfer D N A forms from each D N A strand with T pairing with A via hydrogen bonding.

(a) The DNA molecule is usually a double helix, with the sugar-phosphate backbones of the antiparallel polynucleotide strands (symbolized here by blue ribbons) on the outside of the helix. Hydrogen bonds between pairs of nitrogenous bases hold the two strands together. As illustrated here with symbolic shapes for the bases, adenine (A) can pair only with thymine (T), and guanine (G) can pair only with cytosine (C). Each DNA strand in this figure is the structural equivalent of the polynucleotide diagrammed in Figure 5. 23a. **(b)** A tRNA molecule has a roughly L-shaped structure due to complementary base pairing of antiparallel stretches of RNA. In RNA, A pairs with U.

HHMI BioInteractive Animation

Paired DNA Strands

image

Play

image

Play

**

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In base pairing, only certain bases in the double helix are compatible with each other. Adenine (A) in one strand always pairs with thymine (T) in the other, and guanine (G) always pairs with cytosine (C). Reading the sequence of bases along one strand of the double helix would tell us the sequence of bases along the other strand. If a stretch of one strand has the base sequence 5′-AGGTCCG-3′, then the base-pairing rules tell us that the same stretch of the other strand must have the sequence 3′-TCCAGGC-5′. The two strands of the double helix are *complementary*, each the predictable counterpart of the other. It is this feature of DNA that makes it possible to generate two identical copies of each DNA molecule in a cell that is preparing to divide.

When the cell divides, the copies are distributed to the daughter cells, making them genetically identical to the parent cell. Thus, the structure of DNA accounts for its function of transmitting genetic information whenever a cell reproduces.

RNA molecules, by contrast, exist as single strands. Complementary base pairing can occur, however, between regions of two RNA molecules or even between two stretches of nucleotides in the *same* RNA molecule. In fact, base pairing within an RNA molecule allows it to take on the particular three-dimensional shape necessary for its function. Consider, for example, the type of RNA called *transfer RNA* (*tRNA*), which brings amino acids to the ribosome during the synthesis of a polypeptide. A tRNA molecule is about 80 nucleotides in length. Its functional shape results from base pairing between nucleotides where complementary stretches of the molecule can run antiparallel to each other **(Figure 5.24b)**.

Note that in RNA, adenine (A) pairs with uracil (U); thymine (T) is not present in RNA. Another difference between RNA and DNA is that DNA almost always exists as a double helix, whereas RNA molecules are more variable in shape. RNAs are versatile molecules, and many biologists believe RNA may have preceded DNA as the carrier of genetic information in early forms of life (see Concept 25.1).

Animation

Nucleic Acid Structure

image

Play

image

Play

Concept Check 5.5

- **DRAW IT** In Figure 5.23a, number all the carbons of the top three nucleotides (use primes), circle the nitrogenous bases, and star the phosphates.
- **DRAW IT** A region along one DNA strand has this sequence of nitrogenous bases: 5′-TAGGCCT-3′. Write down its complementary strand, labeling the 5′ and 3′ ends.

*For suggested answers, see*Appendix A.**

Concept 5.6: Genomics and proteomics have transformed biological inquiry and applications

Experimental work in the first half of the 20th century established the role of DNA as the carrier of genetic information, passed from generation to generation, that specified the functioning of living cells and organisms. Once the structure of the DNA molecule was described in 1953, and the linear sequence of nucleotide bases was understood to specify the amino acid sequence of proteins, biologists sought to "decode" genes by learning their nucleotide sequences (often called "base sequences").

The first chemical techniques for *DNA sequencing*, or determining the sequence of nucleotides along a DNA strand, one by one, were developed in the 1970s. Researchers began to study gene sequences, gene by gene, and the more they learned, the more questions they had: How was expression of genes regulated? Genes and their protein products clearly interacted with each other, but how? What was the function, if any, of the DNA that is not part of genes? To fully understand the genetic functioning of a living organism, the entire sequence of the full complement of DNA, the organism's *genome*, would be most enlightening. In spite of the apparent impracticality of this idea, in the late 1980s several prominent biologists put forth an audacious proposal to launch a project that would sequence the entire human genome—all 3 billion bases of it! This endeavor began in 1990 and was effectively completed in the early 2000s.

An unplanned but profound side benefit of this project—the Human Genome Project—was the rapid development of faster and less expensive methods of sequencing. This trend has continued: The cost for sequencing 1 million bases in 2001, well over $5,000, has decreased to less than $0.02 in 2017. And a human genome, the first of which took over 10 years to sequence, could be completed at today's pace in day or less **(Figure 5.25)**. The number of genomes that have been fully sequenced has exploded, generating reams of data and prompting development of ****bioinformatics**, the use of computer software and other computational tools that can handle and analyze these large data sets.

Figure 5.25

Automatic DNA-sequencing machines and abundant computing power enable rapid sequencing of genes and genomes.

Scientists work in a lab.

The reverberations of these developments have transformed the study of biology and related fields. Biologists often look at problems by analyzing large sets of genes or even comparing whole genomes of different species, an approach called **genomics**. A similar analysis of large sets of proteins, including their sequences, is called **proteomics**. (Protein sequences can be determined either by using biochemical

techniques or by translating the DNA sequences that code for them.) These approaches permeate all fields of biology, some examples of which are shown in **Figure 5.26**.

Figure 5.26

AP Make Connections: Contributions of Genomics and Proteomics to Biology

A summary of contributions of genomics and proteomics to many different fields in biology: evolution, conservation biology, paleontology, medical science, and species interactions.

Nucleotide sequencing and the analysis of large sets of genes and proteins can be done rapidly and inexpensively due to advances in technology and information processing. Taken together, genomics and proteomics have advanced our understanding of biology across many different fields.

Evolution: A major aim of evolutionary biology is to understand the relationships among species, both living and extinct. For example, genome sequence comparisons have identified the hippopotamus as the land mammal sharing the most recent common ancestor with whales. (See figure 22.20). Photos: hippopotamus and short-finned pilot whale.

Conservation biology: The tools of molecular genetics and genomics are increasingly used by forensic ecologists to identify which species of animals and plants are killed illegally. In one case, genomic sequences of D N A from illegal shipments of elephant tusks were used to track down poachers and pinpoint the territory where they were operating. Photo: elephants with their tusks in the wild. (See figure 56.9).

Paleontology: New D N A sequencing techniques have allowed decoding of minute quantities of D N A found in ancient tissues from our extinct relatives, the Neanderthals (Homo neanderthalensis). Sequencing the Neanderthal genome has informed our understanding of their physical appearance (as in this reconstruction), as well as their relationship with modern humans. (See figures 34.51 and 34.52). Image: a reconstruction of a Neanderthal man's face, showing a heavy brow ridge, broad nose, and broad mouth.

Medical science: Identifying the genetic basis for human diseases like cancer helps researchers focus their search for potential future treatments. Currently, sequencing the sets of genes expressed in an individual's tumor can allow a more targeted approach to treating cancer, a type of "personalized medicine." (See figures 12.20 and 18.27). Image: two researchers studying a computer screen.

Species interactions: Most plant species exist in a mutually beneficial partnership with fungi (shown at the right) and bacteria associated with the plants' roots; these interactions improve plant growth. Genome sequencing and analysis of gene expression have allowed characterization of plant-associated communities. Such studies will help advance our understanding of such interactions and may improve agricultural practices. (See the chapter 31 scientific skills exercise and figure 37.11). Image: green plant with network of subterranean roots.

MAKE CONNECTIONS *Considering the examples provided here, describe how the approaches of genomics and proteomics help us to address a variety of biological questions.*

*For suggested answer, see**Appendix A.*

HHMI BioInteractive Video

The Making of the Fittest: The Birth and Death of Genes (Icefish)

image

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image

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Perhaps the most significant impact of genomics and proteomics on the field of biology as a whole has been their contributions to our understanding of evolution. In addition to confirming evidence for evolution from the study of fossils and characteristics of currently existing species, genomics has helped us tease out relationships among different groups of organisms that had not been resolved by previous types of evidence, and thus infer evolutionary history.

DNA and Proteins as Tape Measures of Evolution

EVOLUTION We are accustomed to thinking of shared traits, such as hair and milk production in mammals, as evidence of shared ancestry. Because DNA carries heritable information in the form of genes, sequences of genes and their protein products document the hereditary background of an organism. The linear sequences of nucleotides in DNA molecules are passed from parents to offspring; these sequences determine the amino acid sequences of proteins. As a result, siblings have greater similarity in their DNA and proteins than do unrelated individuals of the same species.

Given our evolutionary view of life, we can extend this concept of "molecular genealogy" to relationships between species: We would expect two species that appear to be closely related based on anatomical evidence (and possibly fossil evidence) to also share a greater proportion of their DNA and protein sequences than do less closely related species. In fact, that is the case. An example is the comparison of the β polypeptide chain of human hemoglobin with the corresponding hemoglobin polypeptide in other vertebrates. In this chain of 146 amino acids, humans and gorillas differ in just 1 amino acid, while humans and frogs, more distantly related, differ in 67 amino acids. In the **Scientific Skills Exercise**, you can apply this sort of reasoning to additional species. And this conclusion holds true as well when comparing whole

genomes: The human genome is 95–98% identical to that of the chimpanzee, but only roughly 85% identical to that of the mouse, a more distant evolutionary relative. Molecular biology has added a new tape measure to the toolkit biologists use to assess evolutionary kinship.

Scientific Skills Exercise

Analyzing Polypeptide Sequence Data

AP Science Practices 2, 4, 5, 6

Are Rhesus Monkeys or Gibbons More Closely Related to Humans? In this exercise, you will look at amino acid sequence data for the β polypeptide chain of hemoglobin, often called β-globin. You will then interpret the data to hypothesize whether the monkey or the gibbon is more closely related to humans.

A human baby.

Human

A rhesus monkey.

Rhesus monkey

A gibbon.

Gibbon

How Such Experiments Are Done Researchers can isolate the polypeptide of interest from an organism and then determine the amino acid sequence. More frequently, the DNA of the relevant gene is sequenced, and the amino acid sequence of the polypeptide is deduced from the DNA sequence of its gene.

Data from the Experiments In the data below, the letters give the sequence of the 146 amino acids in β-globin from humans, rhesus monkeys, and gibbons. Because a complete sequence would not fit on one line here, the sequences are divided into three segments: amino acids 1–50, 51–100, and 101–146. The sequences for the three different species are aligned so that you can compare them easily. For example, you can see that for all three species, the first amino acid is V (valine) and the 146th amino acid is H (histidine).

The alignment of amino acid sequences of beta-globin is compared across 3 primates: human, monkey, and gibbon.

Data from Human: http://www.ncbi.nlm.nih.gov/protein/AAA21113.1 ; rhesus monkey: http://www.ncbi.nlm.nih.gov/protein/122634 ; gibbon: http://www.ncbi.nlm.nih.gov/protein/122616

Three sets of sequence alignments of the beta globin gene of human, monkey, and gibbon are compared and the similar sequences are arranged one below the other.

- Human: 1 V H L T P E E K S A V T A L W G K V N V D E V G G E A L G R L L V V Y P W T Q R F F E S F G D L S T
- Monkey: 1 V H L T P E E K N A V T T L W G K V N V D E V G G E A L G R L L L V Y P W T Q R F F E S F G D L S S
- Gibbon: 1 V H L T P E E K S A V T A L W G K V N V D E V G G E A L G R L L V V Y P W T Q R F F E S F G D L S T
- Human: 51 P D A V M G N P K V K A H G K K V L G A F S D G L A H L D N L K G T F A T L S E L H C D K L H V D P
- Monkey: 51 P D A V M G N P K V K A H G K K V L G A F S D G L N H L D N L K G T F A Q L S E L H C D K L H V D P
- Gibbon: 51 P D A V M G N P K V K A H G K K V L G A F S D G L A H L D N L K G T F A Q L S E L H C D K L H V D P
- Human: 101 E N F R L L G N V L V C V L A H H F G K E F T P P V Q A A Y Q K V V A G V A N A L A H K Y H
- Monkey: 101 E N F R L L G N V L V C V L A H H F G K E F T P P V Q A A Y Q K V V A G V A N A L A H K Y H
- Gibbon: 101 E N F R L L G N V L V C V L A H H F G K E F T P P V Q A A Y Q K V V A G V A N A L A H K Y H

INTERPRET THE DATA

- Scan the monkey and gibbon sequences, letter by letter, circling any amino acids that do not match the human sequence.
- How many amino acids differ between the monkey and the human sequences?
- Between the gibbon and human?
- For each nonhuman species, what percent of its amino acids are identical to the human sequence of β-globin?

- Based on these data alone, state a hypothesis for which of these two species is more closely related to humans. What is your reasoning?
- What other evidence could you use to support your hypothesis?

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

Comparing genomic sequences has practical applications as well. In the **Problem-Solving Exercise**, you can see how this type of genomic analysis can help you detect consumer fraud.

Problem-Solving Exercise

Are you a victim of fish fraud?

AP

A grocery store sells wild king salmon and Wild Alaskan Copper River Sockeye salmon, priced at 17.99 and 15.99 dollars respectively.

When buying salmon, perhaps you prefer the more expensive wild-caught Pacific salmon (*Oncorhynchus* species) over farmed Atlantic salmon (*Salmo salar*). But studies reveal that about 40% of the time, you aren't getting the fish you paid for!

In this exercise, you will investigate whether a piece of salmon has been fraudulently labeled.

Your Approach The principle guiding your investigation is that DNA sequences from within a species or from closely related species are more similar to each other than are sequences from more distantly related species.

Your Data You've been sold a piece of salmon labeled as coho salmon (*Oncorhynchus kisutch*). To see whether your fish was labeled correctly, you will compare a short DNA sequence from your sample with known sequences from the same gene for three salmon species. The sequences are as follows:

Sequence alignment of comparison of salmon sequence with other known sequenced.

Data from the International Barcode of Life.

Sample labeled as O. kisutch (coho salmon): 5 prime- C G G C A C C G C C C T A A G T C T C T - 3 prime

Three known sequences of different salmon are listed

- O. kisutch (coho salmon): 5 prime - A G G C A C C G C C C T A A G T C T A C - 3 prime

- O. keta (chum salmon): 5 prime - A G G C A C C G C C C T G A G C C T A C - 3 prime
- Salmo salar (Atlantic salmon): 5 prime - C G G C A C C G C C C T A A G T C T C T - 3 prime

Data from the International Barcode of Life.

Your Analysis

- Circle any bases in the known sequences that do not match the sequence from your fish sample.
- How many bases differ between
- *O. kisutch* and your fish sample?
- *O. keta* and the sample?
- *S. salar* and the sample?
- For each known sequence, what percentage of its bases are identical to your sample?
- Based on these data alone, state a hypothesis for the species identity of your sample. What is your reasoning?

Instructors: A version of this Problem-Solving Exercise can be assigned in Chapter 5 of **Mastering Biology**. A more extensive "Solve It" tutorial is in Chapter 26 of **Mastering Biology**.

Concept Check 5.6

- How would sequencing the entire genome of an organism help scientists to understand how that organism functioned?
- Given the function of DNA, why would you expect two species with very similar traits to also have very similar genomes?

*For suggested answers, see*Appendix A.**